# Chapter 5: Worked Solutions

**SOK-3081 seminar | Exercises 5.1-5.33**

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These solutions follow Sections 5.8.1-5.8.2 of Hill, Griffiths and Lim, *Principles of Econometrics*, 5th edition, pp. 234-247. Exercises 5.1-5.16 use pen and paper and a calculator, with optional R checks. From Exercise 5.17 onward, solutions use R with explanations. R code is displayed rather than executed during PDF compilation; run the blocks in R in their stated order. Numerical results use the supplied observations, the book's rounded output, or the textbook data.

**Notation:** $N$ is the number of observations and $K$ counts all estimated coefficients, including the intercept. Residual degrees of freedom are $N-K$. Exact small-sample inference assumes MR1-MR6. Brief *Instructor notes* suggest questions for the seminar.

## Exercise 5.1: OLS from seven observations

The model is $y_i = x_{i1}\beta_1 + x_{i2}\beta_2 + x_{i3}\beta_3 + e_i$. Table 5.5 has $x_{i1} = 1$ for every observation, so $\beta_1$ is an intercept. There are $N = 7$ observations, $K = 3$ coefficients, and $N - K = 4$ residual degrees of freedom.

### (a) Deviations from the means

Center each variable by subtracting its sample mean: $y_i^* = y_i - \bar{y}$, $x_{i2}^* = x_{i2} - \bar{x}_2$, and $x_{i3}^* = x_{i3} - \bar{x}_3$.

Here the sample means are $\bar{y} = 7/7 = 1$, $\bar{x}_2 = 0/7 = 0$, and $\bar{x}_3 = 0/7 = 0$. Substituting gives $y_i^* = y_i - 1$, $x_{i2}^* = x_{i2} - 0 = x_{i2}$, and $x_{i3}^* = x_{i3} - 0 = x_{i3}$. The original and centered regressor values coincide in this particular sample because both regressor means are zero.

Table 1: Original observations and deviations, based on textbook Table 5.5

| $i$ | $y_i$ | $x_{i2} = x_{i2}^*$ | $x_{i3} = x_{i3}^*$ | $y_i^*$ |
|---|---|---|---|---|
| 1 | 1 | 0 | 1 | 0 |
| 2 | 1 | 1 | -2 | 0 |
| 3 | 4 | 2 | 2 | 3 |
| 4 | 0 | -2 | 1 | -1 |
| 5 | 1 | 1 | -2 | 0 |
| 6 | -2 | -2 | -1 | -3 |
| 7 | 2 | 0 | 1 | 1 |

**Check:** each deviation column sums to zero. The constant column is handled by centering $y$, $x_2$, and $x_3$; we recover the intercept later.

### (b) Cross-products and sums of squares

Multiply the deviations observation by observation, then add each column:

Table 2: Products used in the OLS formulas

| $i$ | $y_i^* x_{i2}^*$ | $(x_{i2}^*)^2$ | $y_i^* x_{i3}^*$ | $x_{i2}^* x_{i3}^*$ | $(x_{i3}^*)^2$ |
|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 1 |
| 2 | 0 | 1 | 0 | -2 | 4 |
| 3 | 6 | 4 | 6 | 4 | 4 |
| 4 | 2 | 4 | -1 | -2 | 1 |
| 5 | 0 | 1 | 0 | -2 | 4 |
| 6 | 6 | 4 | 3 | 2 | 1 |
| 7 | 0 | 0 | 1 | 0 | 1 |
| **Sum** | **14** | **14** | **9** | **0** | **16** |

For compact calculation, write $S_{y2} = 14$, $S_{22} = 14$, $S_{y3} = 9$, $S_{23} = 0$, and $S_{33} = 16$, where each $S$ denotes the corresponding column sum in Table 2.

### (c) Least-squares estimates

Appendix 5A gives the two slope formulas. Their common denominator is $D = S_{22}S_{33} - S_{23}^2 = 14(16) - 0^2 = 224$.

The first slope is $b_2 = (S_{y2}S_{33} - S_{y3}S_{23})/D = [14(16) - 9(0)]/224 = 1$.

The second slope is $b_3 = (S_{y3}S_{22} - S_{y2}S_{23})/D = [9(14) - 14(0)]/224 = 0.5625$.

The intercept is $b_1 = \bar{y} - b_2\bar{x}_2 - b_3\bar{x}_3 = 1 - 1(0) - 0.5625(0) = 1$.

**Fitted model:** $\hat{y}_i = 1 + x_{i2} + 0.5625x_{i3}$.

Here $S_{23} = 0$, so $b_2 = S_{y2}/S_{22}$ and $b_3 = S_{y3}/S_{33}$. This simplification does not hold for multiple regression in general.

*Instructor note:* the regressors are uncorrelated in this sample. This does not establish statistical independence.

### (d) Fitted values and residuals

Calculate $\hat{y}_i = 1 + x_{i2} + 0.5625x_{i3}$, then $\hat{e}_i = y_i - \hat{y}_i$. For observation 1, $\hat{y}_1 = 1.5625$ and $\hat{e}_1 = 1 - 1.5625 = -0.5625$.

Table 3: Fitted values and residuals

| $i$ | Observed $y_i$ | Fitted $\hat{y}_i$ | Residual $\hat{e}_i$ |
|---|---|---|---|
| 1 | 1 | 1.5625 | -0.5625 |
| 2 | 1 | 0.8750 | 0.1250 |
| 3 | 4 | 4.1250 | -0.1250 |
| 4 | 0 | -0.4375 | 0.4375 |
| 5 | 1 | 0.8750 | 0.1250 |
| 6 | -2 | -1.5625 | -0.4375 |
| 7 | 2 | 1.5625 | 0.4375 |

**Check:** $\sum_i \hat{e}_i = 0$. Positive and negative residuals cancel, but that does not mean the fit is perfect.

### (e) Estimate the error variance

The sum of squared residuals is $SSE = (-0.5625)^2 + 3(0.1250)^2 + 3(0.4375)^2 = 0.9375$.

The estimated error variance is $\hat{\sigma}^2 = SSE/(N-K) = 0.9375/4 = 0.234375$.

Use **4**, not 7 or 5, in the denominator: three coefficients have been estimated. The residual standard deviation is $\hat{\sigma} = \sqrt{0.234375} = 0.4841$.

### (f) Correlation between the regressors

The sample correlation is $r_{23} = S_{23}/\sqrt{S_{22}S_{33}} = 0/\sqrt{14(16)} = 0$.

The observations contain no sample linear association between $x_2$ and $x_3$. This concerns the two regressors, not their relationships with $y$.

### (g) Standard error of $b_2$

The estimated variance is $\widehat{\text{var}}(b_2 \mid \mathbf{X}) = \hat{\sigma}^2/[(1-r_{23}^2)S_{22}] = 0.234375/[1(14)] = 0.0167411$.

The standard error is $se(b_2) = \sqrt{0.0167411} = 0.1294$. It measures uncertainty about the slope estimate, whereas $\hat{\sigma} = 0.4841$ measures residual variation in $y$ units.

### (h) SSE, SST, SSR, and $R^2$

We already have $SSE = 0.9375$. The total sum of squares is $SST = \sum_i (y_i - \bar{y})^2 = 0+0+9+1+0+9+1 = 20$.

With an intercept, $SST = SSR + SSE$, so the explained sum of squares is $SSR = 20 - 0.9375 = 19.0625$.

The coefficient of determination is $R^2 = SSR/SST = 19.0625/20 = 1 - 0.9375/20 = 0.953125$.

**Interpretation:** the model explains approximately **95.3%** of the sample variation in $y$ around its mean. This describes in-sample fit; it does not establish causality or guarantee accurate new predictions.

### Exercise 5.2: Inference for $\beta_2$

From Exercise 5.1, $b_2 = 1$, $se(b_2) = 0.1293873$, and $N - K = 4$. The confidence interval and test below use the $t$ distribution with **4 degrees of freedom**, assuming MR1-MR6.

#### (a) A 95% confidence interval

The critical value is $t_{(0.975,4)} = 2.7764$.

The interval is $b_2 \pm t_{(0.975,4)} se(b_2) = 1 \pm 2.7764(0.1293873) = 1 \pm 0.3592$.

**Answer:** the 95% confidence interval is [0.6408, 1.3592].

The interval gives plausible values for the population slope under the model assumptions. Across repeated samples, 95% of intervals from this procedure would contain the fixed true coefficient.

*Instructor note:* ask why we use 2.7764 rather than 1.96. There are only four residual degrees of freedom; treating this as a large sample would understate uncertainty.

#### (b) Test $H_0 : \beta_2 = 1.25$ against $H_1 : \beta_2 \neq 1.25$

Use a **two-sided test**. At the 5% significance level, reject $H_0$ if $|t| \geq 2.7764$.

The test statistic is $t = (b_2 - 1.25)/se(b_2) = (1 - 1.25)/0.1293873 = -1.9322$.

Since $|-1.9322| = 1.9322 < 2.7764$, **do not reject $H_0$ at 5%**. The two-sided p-value is approximately **0.1255**, also above 0.05.

There is insufficient evidence that the population coefficient differs from 1.25. This does not prove that it equals 1.25. The same conclusion follows from part (a): 1.25 lies inside the 95% interval.

*Instructor note:* the estimate is 1, but the null value is 1.25. Testing whether the coefficient equals zero would answer a different question. A coefficient can differ significantly from zero while still being compatible with 1.25.

#### Optional R check for Exercises 5.1-5.2

The calculations above stand on their own. These commands provide a quick check after completing the hand calculations:

```r
d <- data.frame(
  y = c(1, 1, 4, 0, 1, -2, 2),
  x2 = c(0, 1, 2, -2, 1, -2, 0),
  x3 = c(1, -2, 2, 1, -2, -1, 1)
)
model <- lm(y ~ x2 + x3, data = d)
summary(model)
residuals(model)
confint(model, "x2")
t_obs <- (coef(model)["x2"] - 1.25) /
  coef(summary(model))["x2", "Std. Error"]
2 * pt(-abs(t_obs), df = df.residual(model))
```

## Exercise 5.3: Read and interpret regression output

The model is $WALC = \beta_1 + \beta_2 \ln(TOTEXP) + \beta_3 NK + \beta_4 AGE + e$. $WALC$ is the **percentage** of a household's budget spent on alcohol; $TOTEXP$ is total expenditure, $NK$ the number of children, and $AGE$ the household head's age. There are 1,200 London households.

We estimate four coefficients, including the intercept, so $N - K = 1200 - 4 = 1196$. From Table 5.6, $SSE = 46221.62$ and the sample standard deviation of $WALC$ is $s_y = 6.39547$.

### (a) Complete the missing entries

**(i) Intercept t-statistic.** The reported test uses a zero null value: $t = b_1/se(b_1) = 1.4515/2.2019 = 0.6592$.

**(ii) Standard error of $b_2$.** Rearranging $t = b_2/se(b_2)$ gives $se(b_2) = b_2/t = 2.7648/5.7103 = 0.4842$.

**(iii) Estimate $b_3$.** Rearranging the same formula gives $b_3 = t\, se(b_3) = (-3.9376)(0.3695) = -1.4549$.

**(iv) $R^2$.** The reported sample standard deviation uses $N - 1$. Hence $SST = (N-1)s_y^2 = 1199(6.39547)^2 = 49041.54$ and $R^2 = 1 - SSE/SST = 1 - 46221.62/49041.54 = 0.0575$. The model explains about **5.75%** of the sample variation in $WALC$.

**(v) Regression standard error.** The estimated error variance is $\hat{\sigma}^2 = SSE/(N-K) = 46221.62/1196 = 38.6468$. Thus $\hat{\sigma} = \sqrt{38.6468} = 6.2167$ percentage points.

Table 4: Completed output; bold entries were missing from textbook Table 5.6

| Variable | Coefficient | Standard error | t-statistic | Reported p-value |
|---|---|---|---|---|
| Intercept | 1.4515 | 2.2019 | **0.6592** | 0.5099 |
| $\ln(TOTEXP)$ | 2.7648 | **0.4842** | 5.7103 | 0.0000 |
| $NK$ | **-1.4549** | 0.3695 | -3.9376 | 0.0001 |
| $AGE$ | -0.1503 | 0.0235 | -6.4019 | 0.0000 |

**Rounding:** the book reports rounded estimates, standard errors, and t-values. Reconstructing one from the other two can give small discrepancies. Keep the table's reported age t-value rather than replacing it with a ratio of rounded numbers. A printed p-value of 0.0000 means very small, not literally zero.

### (b) Interpret $b_2$, $b_3$, and $b_4$

Hold the other included variables constant in each interpretation.

- **Total expenditure:** this is a linear-log term. A 1% increase in total expenditure corresponds to approximately $2.7648/100 = 0.027648$ **percentage points** more of the budget spent on alcohol. The exact change for a 1% expenditure increase is $2.7648 \ln(1.01) = 0.02751$ percentage points.
- **Children:** one additional child corresponds to about **1.4549 percentage points less** of the budget spent on alcohol.
- **Age:** one additional year of age corresponds to about **0.1503 percentage points less** of the budget spent on alcohol.

These describe the fitted conditional relationships. A causal claim requires the model's exogeneity assumption to be defensible.

*Instructor note:* distinguish **percent** from **percentage points**. A share moving from 6% to 5% falls by 1 percentage point, not by 1%. Do not interpret $b_2 = 2.7648$ as a 2.7648 percentage-point change for a 1% expenditure increase.

### (c) A 95% confidence interval for $\beta_4$

The critical value is $t_{(0.975,1196)} = 1.96195$. The interval is $-0.1503 \pm 1.96195(0.0235) = -0.1503 \pm 0.04611$.

**Answer:** $[-0.1964,\ -0.1042]$ percentage points per year.

Holding expenditure and the number of children constant, an additional year of age is associated with a decrease in the mean budget share of between approximately **0.1042 and 0.1964 percentage points**, using this 95% interval procedure. The whole interval is negative, providing evidence against a zero age coefficient.

### (d) Significance of each coefficient at 5%

Each reported p-value tests $H_0 : \beta_k = 0$ against $H_1 : \beta_k \neq 0$. Reject when $p \leq 0.05$.

Table 5: Two-sided tests against zero

| Coefficient | Evidence from Table 5.6 | Significant at 5%? |
|---|---|---|
| Intercept | $p = 0.5099 > 0.05$ | No |
| $\ln(TOTEXP)$ | Reported $p = 0.0000 < 0.05$ | Yes |
| $NK$ | $p = 0.0001 < 0.05$ | Yes |
| $AGE$ | Reported $p = 0.0000 < 0.05$ | Yes |

The slope coefficients are significant even though $R^2$ is only 0.0575. Statistical evidence of a conditional relationship is different from explaining most of the variation across households.

### (e) Does one additional child reduce the mean share by 2 percentage points?

A decrease of 2 percentage points means $\beta_3 = -2$. The hypotheses are $H_0 : \beta_3 = -2$ and $H_1 : \beta_3 \neq -2$.

Using the reconstructed unrounded estimate $b_3 = -1.4549432$, the test statistic is $t = [b_3 - (-2)]/se(b_3) = (-1.4549432 + 2)/0.3695 = 1.4751$.

At 5%, the two-sided critical values are $\pm 1.96195$. Because $|1.4751| < 1.96195$, **do not reject** $H_0$. The p-value is approximately **0.1404**.

The data do not provide sufficient evidence that the coefficient differs from -2. This does not establish that the effect is exactly -2. Part (d) rejected a coefficient of zero; part (e) tests a different null value, so the conclusions are compatible.

*Instructor note:* ask students to translate the phrase "decreases by 2 percentage points" into the sign of the null value before doing any calculation. Using +2 would test the wrong claim.

## Exercise 5.4: Children, age and the alcohol budget share

The model is $WALC = \beta_1 + \beta_2 \ln(TOTEXP) + \beta_3 NK + \beta_4 AGE + \beta_5 AGE^2 + e$. The dependent variable is the percentage of the household budget spent on alcohol. Changes in $WALC$ are therefore measured in **percentage points**.

Table 5.7 gives the fitted equation $\widehat{WALC} = 8.149 + 2.884 \ln(TOTEXP) - 1.217 NK - 0.5699 AGE + 0.005515 AGE^2$. There are $N = 1200$ observations and $K = 5$ coefficients, so the degrees of freedom are 1195. For a 95% interval, use $t_c = t_{0.975,1195} = 1.96195$.

The relevant estimated covariance matrix is:

| | $b_3$ $(NK)$ | $b_4$ $(AGE)$ | $b_5$ $(AGE^2)$ |
|---|---|---|---|
| $b_3$ | 0.1462 | -0.01774 | 0.0002347 |
| $b_4$ | -0.01774 | 0.03204 | -0.0004138 |
| $b_5$ | 0.0002347 | -0.0004138 | 0.000005438 |

### (a) One additional child

Holding total expenditure and age constant, one additional child changes the mean alcohol budget share by $\beta_3$. The point estimate is $b_3 = -1.217$ **percentage points**.

The standard error is $se(b_3) = \sqrt{0.1462} = 0.38236$. The 95% interval is $-1.217 \pm 1.96195(0.38236)$, or $[-1.9672, -0.4668]$ **percentage points**.

The interval lies below zero: households with one more child have a lower conditional mean alcohol budget share.

### (b) The marginal effect of age at 25, 50 and 75

Because age enters twice, its marginal effect is $\partial E(WALC \mid X)/\partial AGE = \beta_4 + 2AGE\beta_5$. At age $a$, estimate this by $\widehat{m}(a) = -0.5699 + 2a(0.005515)$. The units are percentage points per year, holding expenditure and the number of children constant.

For a linear combination, include the covariance term: $\widehat{\text{var}}[\widehat{m}(a)] = \widehat{\text{var}}(b_4) + 4a^2\widehat{\text{var}}(b_5) + 4a\widehat{\text{cov}}(b_4, b_5)$. Substituting the table gives $0.03204 + 4a^2(0.000005438) + 4a(-0.0004138)$.

For example, at age 25 the estimated effect is $-0.5699 + 50(0.005515) = -0.29415$. Its estimated variance is $0.03204 + 2500(0.000005438) + 100(-0.0004138) = 0.004255$, so its standard error is $\sqrt{0.004255} = 0.06523$. Form each interval as $\widehat{m}(a) \pm t_c\, se[\widehat{m}(a)]$.

| Age | Estimated effect | Standard error | 95% interval |
|---|---|---|---|
| 25 | -0.29415 | 0.06523 | $[-0.4221, -0.1662]$ |
| 50 | -0.01840 | 0.06050 | $[-0.1371, 0.1003]$ |
| 75 | 0.25735 | 0.17394 | $[-0.0839, 0.5986]$ |

Only the interval at age 25 excludes zero. At age 75 the estimated effect is positive, but the interval also permits a small negative effect.

*Instructor note:* the marginal effect is the slope of the age curve at the stated age. Because the curve is quadratic, the exact change from age $a$ to $a+1$ is $b_4 + (2a+1)b_5$, not quite the derivative.

### (c) The age that minimizes the mean alcohol budget share

Set the age slope equal to zero: $\beta_4 + 2a^*\beta_5 = 0$, giving $a^* = -\beta_4/(2\beta_5)$. The fitted curve has positive curvature, $2b_5 = 0.01103 > 0$, so its turning point is a minimum. The estimated minimizing age is $\hat{a}^* = 0.5699/[2(0.005515)] = 51.668$ **years**.

This estimate is a ratio of coefficients. Its standard error needs a **delta-method approximation**: approximate the ratio by a linear function near the coefficient estimates, then use the covariance matrix.

The two relevant derivatives, evaluated at the estimates, are $g_4 = -1/(2b_5)$ and $g_5 = b_4/(2b_5^2)$. The approximate estimated variance is $g_4^2\widehat{\text{var}}(b_4) + g_5^2\widehat{\text{var}}(b_5) + 2g_4g_5\widehat{\text{cov}}(b_4, b_5)$. Using Table 5.7 gives a variance of approximately 37.7112 and a standard error of **6.14095 years**.

An approximate 95% interval is $51.6682 \pm 1.96195(6.14095)$, or [39.62, 63.72] **years**. The use of a $t$ critical value does not make this interval exact. Using 1.96 instead gives almost identical endpoints.

*Instructor note:* the main ideas are finding the turning point and recognizing that a ratio needs an approximate standard error. The derivative calculation can be used as a reference during the seminar.

### (d) What do the results tell us?

More children are associated with a lower alcohol budget share, holding expenditure and age constant. At age 25, the conditional mean share decreases with age. The fitted age curve bottoms out around age 52 and then rises. However, the marginal effects at ages 50 and 75 are not significantly different from zero at 5%.

The minimizing age is uncertain: its approximate interval spans about 40 to 64 years. These are conditional relationships between households; a causal interpretation needs the model's exogeneity assumption. The turning point concerns the mean budget share, not the age at which every individual household spends least on alcohol.

### (e) If the errors are conditionally normal

Under MR1-MR6, the intervals in **(a) and (b) are exact finite-sample $t$ intervals**. Both concern linear combinations of the regression coefficients, even though the age effect changes with the chosen value of age.

The interval in **(c) is a large-sample approximation**. The ratio $-b_4/(2b_5)$ is nonlinear, so its standardized distribution is not exactly $t$, even when the regression errors are normal.

### (f) If the errors are not conditionally normal

None of these intervals is guaranteed to be exact in finite samples. With the other assumptions and suitable large-sample conditions, **all three intervals can be justified approximately**. Normality is needed for the exact finite-sample $t$ result, but not for the usual large-sample normal approximation. The delta method in (c) also requires the population coefficient $\beta_5$ to be nonzero.

## Exercise 5.5: Sample size and slope precision

Both models have an intercept and one explanatory variable. Under MR1-MR6, OLS is unbiased and its conditional slope variance is $\text{var}(b_2 \mid \mathbf{x}) = \sigma^2/S_{xx}$, where $S_{xx} = \sum_{t=1}^{T}(x_t - \bar{x})^2 = \sum x_t^2 - (\sum x_t)^2/T$. Assume $\sigma^2 > 0$ and $T \geq 2$.

The question is whether the accumulated variation in $x_t$ grows without bound as more observations are added.

### (a) A linear time trend: $x_t = t$

The model is $y_t = \beta_1 + \beta_2 t + e_t$, for $t = 1, \ldots, T$. Use the supplied sums: $\sum t = T(T+1)/2$ and $\sum t^2 = T(T+1)(2T+1)/6$.

Subtracting the correction for the mean gives $S_{xx} = T(T+1)(2T+1)/6 - T(T+1)^2/4$. Factoring out $T(T+1)/12$ leaves $S_{xx} = T(T+1)[2(2T+1) - 3(T+1)]/12 = T(T^2-1)/12$.

Therefore $\text{var}(b_2 \mid \mathbf{x}) = 12\sigma^2/[T(T^2-1)] \longrightarrow 0$ as $T \to \infty$.

Since $E(b_2 \mid \mathbf{x}) = \beta_2$, the mean squared error also tends to zero. For any $\varepsilon > 0$, Chebyshev's inequality gives $P(|b_2 - \beta_2| > \varepsilon \mid \mathbf{x}) \leq \text{var}(b_2 \mid \mathbf{x})/\varepsilon^2 \to 0$. Thus $b_2$ **is consistent for** $\beta_2$.

### (b) A shrinking regressor: $x_t = (0.5)^t$

The model is $y_t = \beta_1 + \beta_2(0.5)^t + e_t$. Here $x_t = 2^{-t}$ and $x_t^2 = 4^{-t}$. Using the geometric-series formula gives $\sum x_t = 1 - 2^{-T}$ and $\sum x_t^2 = (1 - 4^{-T})/3$.

Hence $S_{xx} = (1 - 4^{-T})/3 - (1 - 2^{-T})^2/T$, so the slope variance is $\text{var}(b_2 \mid \mathbf{x}) = \sigma^2/[(1 - 4^{-T})/3 - (1 - 2^{-T})^2/T]$.

As $T \to \infty$, the first term in $S_{xx}$ tends to $1/3$ and the second tends to zero. Consequently, $\text{var}(b_2 \mid \mathbf{x}) \to 3\sigma^2 > 0$. The slope does not become arbitrarily precise.

To establish inconsistency here, use the normality assumption: $b_2 \mid \mathbf{x}$ is normal with mean $\beta_2$ and the variance above. Its limiting distribution is therefore $N(\beta_2, 3\sigma^2)$, rather than a point mass at $\beta_2$. There remains a positive probability of an estimation error exceeding any fixed $\varepsilon > 0$. Thus $b_2$ **is not consistent**.

*Instructor note:* a nonzero limiting variance alone does not prove inconsistency for every possible estimator. Here the exact normal distribution makes that conclusion valid.

### (c) Intuition

With $x_t = t$, later observations extend the range of the explanatory variable. The accumulated variation grows without bound, so we can distinguish the slope increasingly well from the intercept.

With $x_t = (0.5)^t$, later observations have explanatory-variable values very close to zero. They help locate the intercept, but add ever less variation for estimating the slope. The accumulated variation approaches a finite limit. **Sample size alone is not enough; the explanatory variable must also contain enough variation.**

## Exercise 5.6: Testing linear combinations of coefficients

The estimates are $b_1 = 2$, $b_2 = 3$ and $b_3 = -1$. The sample has $N = 63$ observations and $K = 3$ coefficients, giving 60 residual degrees of freedom. The estimated covariance matrix is:

| | $b_1$ | $b_2$ | $b_3$ |
|---|---|---|---|
| $b_1$ | 3 | -2 | 1 |
| $b_2$ | -2 | 4 | 0 |
| $b_3$ | 1 | 0 | 3 |

Each alternative is two-sided. For $H_0 : L = L_0$, use $t = (\widehat{L} - L_0)/se(\widehat{L})$ and reject at 5% when $|t| > t_{0.975,60} = 2.00030$.

For a linear combination $\widehat{L} = c_1 b_1 + c_2 b_2 + c_3 b_3$, its estimated variance is the sum of the three terms $c_j^2 \widehat{\text{var}}(b_j)$ and all three covariance terms $2c_j c_k \widehat{\text{cov}}(b_j, b_k)$, for $j < k$. The coefficients $c_j$ determine the signs of the covariance terms.

**(a)** $H_0 : \beta_2 = 0$ **against** $H_1 : \beta_2 \neq 0$

The estimate is $b_2 = 3$ and its standard error is $se(b_2) = \sqrt{4} = 2$. Therefore $t = (3 - 0)/2 = 1.5$.

Since $|1.5| < 2.00030$, **do not reject** $H_0$. The two-sided p-value is approximately **0.1389**. There is insufficient evidence that $\beta_2$ differs from zero.

**(b)** $H_0 : \beta_1 + 2\beta_2 = 5$ **against** $H_1 : \beta_1 + 2\beta_2 \neq 5$

The estimated combination is $\widehat{L} = b_1 + 2b_2 = 2 + 2(3) = 8$. Its estimated variance is $\widehat{\text{var}}(b_1 + 2b_2) = 3 + 2^2(4) + 2(1)(2)(-2) = 11$. Thus $se(\widehat{L}) = \sqrt{11} = 3.31662$ and $t = (8 - 5)/\sqrt{11} = 0.90453$.

Since $|0.90453| < 2.00030$, **do not reject** $H_0$. The two-sided p-value is approximately **0.3693**. The estimated combination of 8 is not significantly different from 5.

**(c)** $H_0 : \beta_1 - \beta_2 + \beta_3 = 4$ **against** $H_1 : \beta_1 - \beta_2 + \beta_3 \neq 4$

The estimated combination is $\widehat{L} = b_1 - b_2 + b_3 = 2 - 3 - 1 = -2$. The weights are $(c_1, c_2, c_3) = (1, -1, 1)$. Its estimated variance is $3 + (-1)^2(4) + 3 + 2(1)(-1)(-2) + 2(1)(1)(1) + 2(-1)(1)(0) = 3 + 4 + 3 + 4 + 2 + 0 = 16$.

Thus $se(\widehat{L}) = \sqrt{16} = 4$ and $t = (-2 - 4)/4 = -1.5$. Since $|-1.5| < 2.00030$, **do not reject** $H_0$. The two-sided p-value is approximately **0.1389**. There is insufficient evidence that this combination differs from 4.

*Instructor note:* the covariance between $b_1$ and $b_2$ is negative, but it contributes positively to the variance in (c), because $b_2$ has a negative weight. Do not add standard errors or ignore covariances. Failure to reject any of these hypotheses does not establish that it is true.

**Optional R check:** obtain the three two-sided p-values with `2 * pt(-abs(c(1.5, 3 / sqrt(11), -1.5)), df = 60)`.

## Exercise 5.7: Standard errors and tests from summary information

The model is $y = \beta_1 + \beta_2 x_2 + \beta_3 x_3 + e$. There are $N = 203$ observations and $K = 3$ coefficients, so all tests use 200 residual degrees of freedom. The estimates are $b_2 = 0.7176$ and $b_3 = 1.0516$.

### (a) Standard errors of the two slopes

First estimate the error variance: $\hat{\sigma}^2 = SSE/(N-K) = 6800/200 = 34$. The correlation between the explanatory variables is $r_{23} = 0.7087$, so $1 - r_{23}^2 = 1 - 0.7087^2 = 0.49774431$.

With two explanatory variables, the slope standard errors are $se(b_j) = \sqrt{\hat{\sigma}^2/[S_{jj}(1-r_{23}^2)]}$, where $S_{jj} = \sum_i (x_{ij} - \bar{x}_j)^2$. Thus $se(b_2) = \sqrt{34/[1780.7(0.49774431)]} = 0.19586$ and $se(b_3) = \sqrt{34/[3453.3(0.49774431)]} = 0.14064$.

*Instructor note:* $r_{23}$ measures how closely the two explanatory variables move together. Its square belongs in the denominator. Holding the error variance and each variable's total variation fixed, greater correlation increases the slope standard errors.

### (b) Is $\beta_2$ different from zero at 5%?

Test $H_0 : \beta_2 = 0$ against $H_1 : \beta_2 \neq 0$. The statistic is $t = (0.7176 - 0)/0.19586 = 3.66388$. The two-sided critical values are $\pm t_{0.975,200} = \pm 1.97190$.

Since $3.66388 > 1.97190$, **reject** $H_0$. The p-value is approximately **0.00032**. There is evidence of a nonzero conditional relationship between $y$ and $x_2$.

### (c) Is $\beta_3$ greater than 0.9 at 10%?

Test $H_0 : \beta_3 \leq 0.9$ against $H_1 : \beta_3 > 0.9$. Calculate at the null boundary: $t = (1.0516 - 0.9)/0.14064 = 1.07790$. This is an upper-tail test, so the critical value is $t_{0.90,200} = 1.28580$.

Since $1.07790 < 1.28580$, **do not reject** $H_0$. The one-sided p-value is approximately **0.1412**. Although the point estimate exceeds 0.9, the evidence is not strong enough to conclude that the population coefficient exceeds 0.9.

### (d) Are the slopes equal at 1%?

Rewrite $H_0 : \beta_2 = \beta_3$ as $H_0 : \beta_2 - \beta_3 = 0$, against $H_1 : \beta_2 - \beta_3 \neq 0$. The estimated difference is $b_2 - b_3 = 0.7176 - 1.0516 = -0.3340$.

Its estimated variance is $\widehat{\text{var}}(b_2 - b_3) = se(b_2)^2 + se(b_3)^2 - 2\widehat{\text{cov}}(b_2, b_3) = 0.1958578^2 + 0.1406433^2 - 2(-0.019521) = 0.0971828$. Hence $se(b_2 - b_3) = 0.31174$ and $t = -0.3340/0.31174 = -1.07140$.

The two-sided critical values are $\pm t_{0.995,200} = \pm 2.60063$. Since $|-1.07140| < 2.60063$, **do not reject equal slopes**. The p-value is approximately **0.2853**.

*Instructor note:* subtracting two coefficients does not mean subtracting their variances. Here the covariance is negative, so the term $-2\widehat{\text{cov}}(b_2, b_3)$ increases the variance of the difference. Use unrounded standard errors in calculations.

## Exercise 5.8: Population, GDP and Olympic medals in 1996

The sample contains the 79 countries that won at least one medal. The estimated model is $\widehat{MEDALS} = 5.917 + 0.01813POPM + 0.01026GDPB$, with $R^2 = 0.4879$. Population is in millions and GDP is in billions of 1995 dollars. The reported slope standard errors are $se(b_2) = 0.00819$ and $se(b_3) = 0.00136$. With $K = 3$, the tests use $79 - 3 = 76$ degrees of freedom. Assume MR1-MR6 as stated in the question.

### (a) Interpret the slope estimates

Holding GDP constant, an additional **one million people** is associated with an increase of **0.01813 in the expected number of medals**. For example, 100 million more people corresponds to $100(0.01813) = 1.813$ more expected medals.

Holding population constant, an additional **one billion dollars of GDP** is associated with an increase of **0.01026 expected medals**. An additional 100 billion dollars corresponds to $100(0.01026) = 1.026$ more expected medals.

The dependent variable is a count, but its conditional mean need not be an integer. The estimates describe the medal-winning countries in this sample; extending them to countries with no medals requires further justification.

### (b) Interpret $R^2$

The model explains **48.79% of the sample variation in medal counts around their sample mean**. Before fitting the model, countries' medal counts already differed from the mean. Population and GDP explain about half of that variation in this sample; 51.21% remains in the residuals.

This is not the percentage of medals correctly predicted or the percentage of medals caused by population and GDP.

### (c) A positive GDP relationship: 1% and 5% tests

Test $H_0 : \beta_3 = 0$ against $H_1 : \beta_3 > 0$. The upper-tail statistic is $t = 0.01026/0.00136 = 7.54412$. At 1%, the critical value is $t_{0.99,76} = 2.37642$. Since $7.54412 > 2.37642$, **reject $H_0$ at 1%**.

At 5%, the critical value falls to $t_{0.95,76} = 1.66515$, so **reject at 5% as well**. The one-sided p-value is about $4.09 \times 10^{-11}$, far below either significance level. There is strong evidence of a positive GDP coefficient, controlling for population.

### (d) A positive population relationship: 1% and 5% tests

Test $H_0 : \beta_2 = 0$ against $H_1 : \beta_2 > 0$. The statistic is $t = 0.01813/0.00819 = 2.21368$.

At 1%, $2.21368 < 2.37642$, so **do not reject $H_0$**. At 5%, $2.21368 > 1.66515$, so **reject $H_0$**. The one-sided p-value is **0.01493**, between 0.01 and 0.05.

*Instructor note:* the data and test statistic stay the same when the significance level changes. Only the threshold for rejecting the null changes. These are one-sided tests, so do not use the two-sided 5% critical value.

### (e) Four different null values for the population coefficient

Each test has the form $H_0 : \beta_2 = \beta_{2,0}$ against $H_1 : \beta_2 \neq \beta_{2,0}$. Use $t = (0.01813 - \beta_{2,0})/0.00819$. At 5%, the two-sided critical values are $\pm t_{0.975,76} = \pm 1.99167$.

| Part | Null value | $t$ statistic | Two-sided p-value | Decision at 5% |
|---|---|---|---|---|
| (i) | 0.01 | 0.99267 | 0.3240 | Do not reject |
| (ii) | 0.02 | -0.22833 | 0.8200 | Do not reject |
| (iii) | 0.03 | -1.44933 | 0.1514 | Do not reject |
| (iv) | 0.04 | -2.67033 | 0.0093 | Reject |

For example, the last statistic is $t = (0.01813 - 0.04)/0.00819 = -2.67033$. Its magnitude exceeds 1.99167, so 0.04 is rejected.

**These results are not contradictory.** Failure to reject 0.01, 0.02 and 0.03 does not establish that the coefficient equals all three numbers. It means the sample is not precise enough to rule out any of those values at the stated level. A null value farther from the estimate can be rejected.

### (f) A 95% confidence interval for $\beta_2$

The interval is $b_2 \pm t_c \, se(b_2) = 0.01813 \pm 1.99167(0.00819)$, giving [0.001818, 0.034442] **medals per additional million people**, holding GDP constant.

All three nonrejected null values in (e) lie inside this interval; 0.04 lies outside. This illustrates the connection between a two-sided 5% test and a 95% confidence interval.

The interval is entirely positive but relatively wide compared with the point estimate. It supports a positive population coefficient in a two-sided 5% test, while leaving substantial uncertainty about the size of that coefficient. This is compatible with the one-sided 1% test in (d) failing to reject zero: that test uses a stricter significance level.

*Instructor note:* draw a number line and mark the confidence interval, the point estimate and the four null values. Ask students to predict the decisions before calculating the four test statistics.

## Exercise 5.9: A level-log model for Olympic medals in 1992

The model uses 63 medal-winning countries, excluding the United Kingdom. The fitted equation is $\widehat{MEDALS} = -13.153 + 2.764\ln(POPM) + 4.270\ln(GDPB)$, with $R^2 = 0.275$. The slope standard errors are 2.070 and 1.718, respectively. With $N = 63$ and $K = 3$, use 60 degrees of freedom. Assume MR1-MR6 as stated in the question.

### (a) Interpret the slope estimates

This is a **level-log model**: medals are in levels, while population and GDP enter as natural logarithms. A 1% increase in an explanatory variable changes its logarithm by approximately 0.01.

Holding GDP constant, a **1% increase in population** is associated with approximately $2.764/100 = \mathbf{0.02764}$ additional expected medals. Holding population constant, a **1% increase in GDP** is associated with approximately $4.270/100 = \mathbf{0.04270}$ additional expected medals.

For a finite percentage increase of $p$%, the exact fitted change is $b_j \ln(1 + p/100)$ medals. For example, a 10% GDP increase corresponds to $4.270\ln(1.10) = 0.40697$ more expected medals.

*Instructor note:* the change in expected medals is in medal units, not percent. These coefficients are not elasticities because the dependent variable is not logged.

### (b) Interpret $R^2$

Log population and log GDP explain **27.5% of the sample variation in medal counts around their sample mean**. The remaining 72.5% is unexplained by this fitted model.

Do not interpret the lower $R^2$ than in Exercise 5.8 as a clean comparison of functional forms: both the Olympic year and the sample differ between the exercises.

### (c) A positive GDP relationship: 10% and 5% tests

Test $H_0 : \beta_3 = 0$ against $H_1 : \beta_3 > 0$. The statistic is $t = 4.270/1.718 = 2.48545$. The upper-tail critical values are $t_{0.90,60} = 1.29582$ at 10% and $t_{0.95,60} = 1.67065$ at 5%.

Since 2.48545 exceeds both, **reject $H_0$ at both levels**. The one-sided p-value is **0.00787**. There is evidence of a positive GDP coefficient, holding population constant.

### (d) A positive population relationship: 10% and 5% tests

Test $H_0 : \beta_2 = 0$ against $H_1 : \beta_2 > 0$. The statistic is $t = 2.764/2.070 = 1.33527$.

At 10%, $1.33527 > 1.29582$, so **reject $H_0$**. At 5%, $1.33527 < 1.67065$, so **do not reject $H_0$**. The one-sided p-value is **0.09342**. The evidence of a positive population coefficient is sufficient at 10%, but not at 5%.

### (e) Expected medals for the United Kingdom

The United Kingdom has $POPM = 58$ and $GDPB = 1010$. Use these values in their stated units before taking natural logs: $\ln(58) = 4.060443$ and $\ln(1010) = 6.917706$.

The estimated conditional mean is $\widehat{MEDALS}_0 = -13.153 + 2.764\ln(58) + 4.270\ln(1010) =$ **27.60867** medals. The question supplies the standard error of this estimated mean: $se(\widehat{MEDALS}_0) = 4.22196$.

The 95% confidence interval is $27.60867 \pm t_{0.975,60}(4.22196) = 27.60867 \pm 2.00030(4.22196)$, or [19.16349, 36.05384] **expected medals**.

This is an interval for the **conditional mean**, not a prediction interval for the United Kingdom's actual medal count. A prediction interval also allows for the new observation's regression error and would be wider.

### (f) Compare the estimate with the United Kingdom's 20 medals

The point estimate overpredicts the observed count by $27.60867 - 20 = 7.60867$ medals. The value 20 lies inside the confidence interval from (e).

For the hypothesis specified in the question, define $L = \beta_1 + \ln(58)\beta_2 + \ln(1010)\beta_3$ and test $H_0 : L = 20$ against $H_1 : L \neq 20$. The statistic is $t = (27.60867 - 20)/4.22196 = 1.80216$. The two-sided p-value is approximately **0.07654**.

At 5%, **do not reject** $H_0$, because $0.07654 > 0.05$. At 10%, it would be rejected. The model's estimated mean is not significantly different from 20 at 5%, but the interval is fairly wide and the point prediction misses the observed count by about eight medals. This one comparison does not establish that the model predicts well.

*Instructor note:* the exercise's test treats 20 as a hypothesized conditional mean. The 20 medals actually won are a single observed outcome, not a known population mean. Evaluating the uncertainty of that outcome would use a prediction interval instead.

### (g) Standard error of the estimated mean: the expression

Let $a = \ln(58)$ and $c = \ln(1010)$. The estimated mean is the linear combination $\widehat{L} = b_1 + ab_2 + cb_3$. Its estimated variance contains three variance terms and three covariance terms:

$$\begin{aligned}\widehat{\mathrm{var}}(\widehat{L}) = \widehat{\mathrm{var}}(b_1) + a^2\widehat{\mathrm{var}}(b_2) + c^2\widehat{\mathrm{var}}(b_3) \\ + 2a\widehat{\mathrm{cov}}(b_1, b_2) + 2c\widehat{\mathrm{cov}}(b_1, b_3) + 2ac\widehat{\mathrm{cov}}(b_2, b_3).\end{aligned}$$

The requested standard error is the square root: $se(\widehat{L}) = \sqrt{\widehat{\mathrm{var}}(\widehat{L})}$. The three reported coefficient standard errors alone are not enough to reconstruct 4.22196: the covariances are also needed.

**Optional R check:** the p-value in (f) is `2 * pt(-abs((27.60867 - 20) / 4.22196), df = 60)`.

## Exercise 5.10: Wheat yield, time and rainfall

For 1950-1996, the fitted model is $\widehat{YIELD}_t = 0.1717 + 0.01117t + 0.05238RAIN_t$. Yield is in tonnes per hectare and rainfall is in inches during the growing season. The trend is $t = 1$ in 1950 and $t = 47$ in 1996. The slope standard errors are 0.00262 and 0.01367, respectively.

There are $T = 47$ observations and $K = 3$ coefficients, giving 44 residual degrees of freedom. The question specifies MR1-MR5. We use conventional $t$ inference below; without the normality assumption MR6, these tests are approximate rather than exact.

### (a) Interpret the trend and rainfall coefficients

Holding rainfall constant, one more year is associated with **0.01117 additional tonnes per hectare in mean yield**. The trend is intended to capture technological progress, although it could also capture other gradual changes omitted from the model.

Holding the trend constant, one additional inch of rainfall is associated with **0.05238 additional tonnes per hectare in mean yield**. This linear specification assumes the same rainfall slope at every rainfall level.

### (b) Does the annual trend increase exceed 0.01?

Test $H_0 : \beta_2 \leq 0.01$ against $H_1 : \beta_2 > 0.01$. At the null boundary, $t = (0.01117 - 0.01)/0.00262 = 0.44656$. The upper-tail 5% critical value is $t_{0.95,44} = 1.68023$.

Since $0.44656 < 1.68023$, **do not reject** $H_0$. The one-sided p-value is **0.3287**. The estimated trend exceeds 0.01, but there is insufficient evidence that the population trend does.

### (c) Is the rainfall effect different from 0.03?

Test $H_0 : \beta_3 = 0.03$ against $H_1 : \beta_3 \neq 0.03$. The statistic is $t = (0.05238 - 0.03)/0.01367 = 1.63716$. The two-sided 5% critical values are $\pm t_{0.975,44} = \pm 2.01537$.

Since $|1.63716| < 2.01537$, **do not reject** $H_0$. The two-sided p-value is **0.1087**. An effect of 0.03 tonnes per hectare per additional inch is compatible with these data at 5%.

### (d) Why add squared rainfall?

The new fitted equation is $\widehat{YIELD}_t = -0.6759 + 0.011671t + 0.2229RAIN_t - 0.008155RAIN_t^2$. Its marginal rainfall effect is $\beta_3 + 2\beta_4 RAIN$.

The quadratic allows the effect of rainfall to diminish: additional rain can help when conditions are dry, while too much rain can harm the crop. The **negative squared-term coefficient has the expected sign** for this pattern. The rainfall slope is no longer simply 0.2229.

### (e) Repeat the trend test with squared rainfall included

There are now $K = 4$ coefficients, so use $47 - 4 = 43$ degrees of freedom. The trend estimate is 0.011671 with standard error 0.00250. For the same upper-tail test as in (b), $t = (0.011671 - 0.01)/0.00250 = 0.66840$.

Since $0.66840 < t_{0.95,43} = 1.68107$, **do not reject** $H_0$. The one-sided p-value is **0.2537**. The conclusion is unchanged.

### (f) Test the rainfall effect at mean rainfall

At $r = \overline{RAIN} = 10.059$, the marginal effect is $m(r) = \beta_3 + 2r\beta_4$. Its estimate is $\widehat{m} = 0.2229 + 2(10.059)(-0.008155) = \mathbf{0.05883771}$ tonnes per hectare per additional inch.

The estimated variance is $\widehat{\text{var}}(\widehat{m}) = se(b_3)^2 + 4r^2 se(b_4)^2 + 4r\widehat{\text{cov}}(b_3, b_4)$. Substitute the supplied values: $0.0734^2 + 4(10.059)^2(0.003453)^2 + 4(10.059)(-0.0002493) = 0.0001824524$. Thus $se(\widehat{m}) = 0.01350749$.

Test $H_0 : m(10.059) = 0.03$ against $H_1 : m(10.059) \neq 0.03$. The statistic is $t = (0.05883771 - 0.03)/0.01350749 = 2.13494$. Since $|2.13494| > t_{0.975,43} = 2.01669$, **reject $H_0$ at 5%**. The two-sided p-value is **0.03851**.

*Instructor note:* the null now concerns a combination of two coefficients at a particular rainfall level. Testing $\beta_3 = 0.03$ would test the rainfall slope at zero rainfall, not at the mean. Keep the covariance term and avoid early rounding: the variance calculation subtracts numbers that are close in size.

### (g) Forecast 1997 yield when rainfall is 9.48 inches

Since 1996 corresponds to $t = 47$, use $t = 48$ for 1997. The quadratic model gives $\widehat{YIELD}_{1997} = -0.6759 + 0.011671(48) + 0.2229(9.48) - 0.008155(9.48)^2 = \mathbf{1.26451}$ tonnes per hectare.

This is a point forecast conditional on the stated rainfall.

### (h) Forecast before rainfall has been observed

We need an assumption about the unknown rainfall. A natural choice is to use the historical mean and variance as estimates of its 1997 distribution: $E(RAIN) \approx 10.059$ and $\text{var}(RAIN) \approx 2.624^2$.

**Linear model:** because the fitted equation is linear in rainfall, substituting expected rainfall also gives expected fitted yield: $0.1717 + 0.01117(48) + 0.05238(10.059) = \mathbf{1.23475}$ tonnes per hectare.

**Quadratic model:** expected squared rainfall is not the square of expected rainfall. Use $E(RAIN^2) = \text{var}(RAIN) + [E(RAIN)]^2 \approx 2.624^2 + 10.059^2$. The forecast of expected yield is therefore $-0.6759 + 0.011671(48) + 0.2229(10.059) - 0.008155(10.059^2 + 2.624^2) = \mathbf{1.24516}$ tonnes per hectare.

Simply inserting mean rainfall into both rainfall terms gives **1.30131** tonnes per hectare. That is the fitted yield *at mean rainfall*, rather than the average over uncertain rainfall. The latter is lower by $0.008155(2.624^2) = 0.05615$ because the curve is concave.

*Instructor note:* if the historical empirical distribution is used instead, its second moment is $\bar{r}^2 + (46/47)s_r^2$, giving 1.24635. The small difference comes from using the sample variance to estimate a population variance versus averaging directly over the 47 historical rainfall observations. State the forecasting assumption.

## Exercise 5.11: Education, experience and wages

The model is $WAGE = \beta_1 + \beta_2 EDUC + \beta_3 EXPER + \beta_4 EXPER^2 + e$. The fitted equation from 600 observations is $\widehat{WAGE} = -16.308 + 2.329 EDUC + 0.5240 EXPER - 0.007582 EXPER^2$. Use $600 - 4 = 596$ degrees of freedom and $t_c = t_{0.975,596} = 1.96395$ for the intervals below. The linear-combination intervals use conventional $t$ inference; they are exact under MR1-MR6 and approximate under suitable large-sample conditions without normality.

### (a) The marginal effect of experience

Holding education constant, the experience slope is $\partial E(WAGE \mid X)/\partial EXPER = \beta_3 + 2\beta_4 EXPER$. Both the linear and squared experience terms contribute.

### (b) Expected signs

We expect $\beta_2 > 0$ because more education raises expected wages, holding experience constant; $\beta_3 > 0$ because initial experience raises expected wages; and $\beta_4 < 0$ because the experience gain diminishes and eventually becomes negative. The estimated slopes have these signs.

### (c) When does mean wage start to decline?

Set the experience slope to zero: $\beta_3 + 2\beta_4 E^* = 0$, giving $E^* = -\beta_3/(2\beta_4)$. With $\beta_3 > 0$ and $\beta_4 < 0$, this is a maximum: mean wage rises before $E^*$ and declines after it. The fitted turning point is $-0.5240/[2(-0.007582)] = 34.55553$ years of experience.

### (d) Four 95% confidence intervals

**(i) Education.** The marginal effect is $\beta_2$, with $se(b_2) = 0.163$. Its interval is $2.329 \pm 1.96395(0.163)$, or [2.00888, 2.64912] **wage units per additional year of education**.

**(ii)-(iii) Experience at 4 and 25 years.** At experience $a$, the estimated effect is $\widehat{m}(a) = 0.5240 - 0.015164a$. Its estimated variance is $0.1263^2 + 4a^2(0.002532)^2 + 4a(-0.00030526)$. Take the square root for its standard error and use $\widehat{m}(a) \pm t_c\, se[\widehat{m}(a)]$.

| Experience | Estimated effect | Standard error | 95% interval |
|---|---|---|---|
| 4 | 0.463344 | 0.107135 | [0.25294, 0.67375] |
| 25 | 0.144900 | 0.038122 | [0.07003, 0.21977] |

Both intervals are positive: expected wage is still increasing at these experience levels, but the estimated gain is smaller at 25. The units are wage units per additional year of experience.

**(iv) Experience at the wage maximum.** The ratio $\widehat{E^*} = -b_3/(2b_4)$ needs a delta-method standard error, as in Exercise 5.4(c). The derivatives are $g_3 = -1/(2b_4)$ and $g_4 = b_3/(2b_4^2)$. Its approximate estimated variance is $g_3^2(0.1263)^2 + g_4^2(0.002532)^2 + 2g_3g_4(-0.00030526)$, giving $se(\widehat{E^*}) = 4.36400$ **years**. The approximate interval is $34.55553 \pm 1.96395(4.36400)$, or [25.98484, 43.12622] **years of experience**.

*Instructor note:* the ratio interval is approximate even with normal errors. The turning point concerns the conditional mean wage curve, not a known peak in every individual worker's career.

## Exercise 5.12: Controlling for house age with residuals

The target model is $PRICE = \beta_1 + \beta_2 SQFT + \beta_3 AGE + e$, using 850 house sales. Price is in thousands of dollars and size is in hundreds of square feet.

The preliminary regressions remove a constant and age from both variables: $PRICE = \alpha_1 + \alpha_2 AGE + v$ and $SQFT = \delta_1 + \delta_2 AGE + u$. Their residuals satisfy $\sum \hat{v}_i^2 = 10377817$, $\sum \hat{u}_i^2 = 75773.4$ and $\sum \hat{u}_i \hat{v}_i = 688318$.

### (a) Estimate the size coefficient in the multiple regression

By the **Frisch-Waugh-Lovell theorem**, regress the price residuals $\hat{v}_i$ on the size residuals $\hat{u}_i$. The slope is the same as the size coefficient in the full model: $b_2 = \sum \hat{u}_i \hat{v}_i / \sum \hat{u}_i^2 = 688318/75773.4 = 9.08390$.

Holding age constant, another 100 square feet is associated with about **9.084 thousand dollars** in mean selling price, equivalent to about 90.84 dollars per square foot.

*Instructor note:* $\hat{u}_i$ is the part of size not explained by age; $\hat{v}_i$ is the part of price not explained by age. Their relationship isolates the size slope after controlling for age. Both preliminary regressions include intercepts, so both residual series have mean zero.

### (b) Show the residual sum-of-squares identity

The full-model residual is $\hat{e}_i = \hat{v}_i - b_2 \hat{u}_i$. Expanding and summing gives $\sum \hat{e}_i^2 = \sum \hat{v}_i^2 - 2b_2 \sum \hat{u}_i \hat{v}_i + b_2^2 \sum \hat{u}_i^2$.

From (a), $b_2 \sum \hat{u}_i^2 = \sum \hat{u}_i \hat{v}_i$, so the final term equals $b_2 \sum \hat{u}_i \hat{v}_i$. Combining terms gives the requested identity: $\sum \hat{e}_i^2 = \sum \hat{v}_i^2 - b_2 \sum \hat{u}_i \hat{v}_i$.

### (c) Estimate the error variance

Using the unrounded slope, $SSE = 10377817 - (688318/75773.4)(688318) = 4125205.0132$. The full regression has three coefficients, so $\hat{\sigma}^2 = SSE/(850-3) = 4125205.0132/847 = 4870.3719$.

### (d) The standard error of the size coefficient

The size variation remaining after controlling for age is $\sum \hat{u}_i^2$. Therefore $se(b_2) = \sqrt{\hat{\sigma}^2 / \sum \hat{u}_i^2} = \sqrt{4870.3719/75773.4} = 0.253526$. Use the full model's 847 residual degrees of freedom, even when calculating the slope from the residual regression.

### (e) Is the size coefficient below 9.5?

Test $H_0 : \beta_2 \geq 9.5$ against $H_1 : \beta_2 < 9.5$. At the boundary, $t = (9.0839002 - 9.5)/0.2535261 = -1.64125$. The lower-tail p-value is $P(t_{847} \leq -1.64125) = 0.05056$.

At 5%, **do not reject** $H_0$, narrowly. At 10%, reject it. The estimate is below 9.5, but the evidence just misses the 5% threshold. Retain unrounded values rather than rounding the statistic to -1.65 or the p-value to 0.05 before deciding.

**Optional R check:** `pt(-1.641251, df = 847)` gives the lower-tail p-value.

## Exercise 5.13: Okun's Law and unemployment forecasts

Okun's Law is $U_t - U_{t-1} = -\gamma(G_t - G_N)$. $G_N$ is the quarterly growth rate associated with unchanged unemployment, while $0 < \gamma < 1$ is an adjustment coefficient. Growth rates are entered in percent; changes in unemployment are in percentage points.

### (a) Rewrite the model as a regression

Define $DU_t = U_t - U_{t-1}$ and expand the right-hand side: $DU_t = \gamma G_N - \gamma G_t = \beta_1 + \beta_2 G_t$. Thus $\beta_1 = \gamma G_N$ and $\beta_2 = -\gamma$. For estimation, add an error: $DU_t = \beta_1 + \beta_2 G_t + e_t$.

### (b) Estimate the adjustment coefficient and normal growth rate

The fitted equation is $\widehat{DU}_t = 0.1989 - 0.2713 G_t$, with $\hat{\sigma} = 0.2749$. There are 180 quarters from 1970 Q1 through 2014 Q4, so the residual degrees of freedom are $180 - 2 = 178$ and $t_{0.975,178} = 1.97338$. We use conventional regression inference under the maintained error assumptions; exact finite-sample $t$ results additionally require MR6.

The adjustment estimate is $\hat{\gamma} = -b_2 = 0.2713$. Growth one percentage point above normal is associated with a 0.2713 percentage-point fall in unemployment over the quarter.

The normal growth estimate is $\widehat{G}_N = -b_1/b_2 = -0.1989/(-0.2713) = 0.73314\%$ **per quarter**. At this growth rate, the predicted unemployment change is zero.

### (c) Standard errors and significance against zero

The covariance matrix supplies $\widehat{\text{var}}(b_1) = 0.0007212$, $\widehat{\text{var}}(b_2) = 0.0006113$, and $\widehat{\text{cov}}(b_1, b_2) = -0.0004277$. Take square roots of the diagonal entries to obtain $se(b_1) = 0.026855$ and $se(b_2) = 0.024724$. Because $\hat{\gamma} = -b_2$, $se(\hat{\gamma}) = se(b_2) = 0.024724$.

For $\widehat{G}_N = -b_1/b_2$, use the delta method. The derivatives are $g_1 = -1/b_2$ and $g_2 = b_1/b_2^2$. The approximate estimated variance is $g_1^2(0.0007212) + g_2^2(0.0006113) + 2g_1 g_2(-0.0004277)$, giving $se(\widehat{G}_N) = 0.075777$ percentage points of growth.

| Quantity | Estimate | Standard error | Statistic against zero |
|---|---|---|---|
| $\beta_1$ | 0.198900 | 0.026855 | 7.4064 |
| $\beta_2$ | -0.271300 | 0.024724 | -10.9729 |
| $\gamma$ | 0.271300 | 0.024724 | 10.9729 |
| $G_N$ | 0.733137 | 0.075777 | 9.6749 |

All four magnitudes exceed 1.97338, so **all are significantly different from zero at 5%** using these procedures. The ratio-based statistic for $G_N$ is approximate even under normal errors.

### (d) Is normal growth 0.8% per quarter?

Test $H_0 : G_N = 0.8$ against $H_1 : G_N \neq 0.8$. Using the delta-method standard error from (c), $t = (0.73313675 - 0.8)/0.07577686 = -0.88237$. Its approximate two-sided p-value is **0.3788**. Since $|-0.88237| < 1.97338$, **do not reject $H_0$ at 5%.**

The same null can be written as $\beta_1 + 0.8\beta_2 = 0$, because $G_N = -\beta_1/\beta_2$ and $\beta_2 \neq 0$. Testing this linear restriction gives an estimate of $-0.01814$. Its estimated variance is $0.0007212 + 0.8^2(0.0006113) + 2(0.8)(-0.0004277) = 0.000428112$, so its standard error is $\sqrt{0.000428112} = 0.020691$ and $t = -0.87672$. Its p-value is **0.3818**, giving the same decision. The small difference arises from approximating the nonlinear ratio in the first test.

### (e) A 95% interval for the adjustment coefficient

The interval is $\hat{\gamma} \pm t_c\, se(\hat{\gamma}) = 0.2713 \pm 1.97338(0.02472448)$, or [0.22251, 0.32009]. The entire interval lies in the economically proposed range $(0, 1)$.

### (f) A 95% interval for expected unemployment in 2015 Q1

The previous unemployment rate is 5.7991 and the growth input is $G_{2015Q1} = 0.062$. Enter 0.062 directly in the fitted equation; do not multiply it by 100 or divide it by 100.

First predict the change: $\widehat{DU}_0 = 0.1989 - 0.2713(0.062) = 0.1820794$ percentage points. Then add the known previous unemployment rate: $\widehat{U}_0 = 5.7991 + 0.1820794 = 5.9811794\%$.

The fixed value 5.7991 adds no estimation variance. Therefore the standard error of the estimated conditional mean is $se(\widehat{U}_0) = \sqrt{\widehat{\text{var}}(b_1) + 0.062^2\widehat{\text{var}}(b_2) + 2(0.062)\widehat{\text{cov}}(b_1, b_2)}$. Substituting gives $\sqrt{0.0007212 + 0.062^2(0.0006113) + 2(0.062)(-0.0004277)} = 0.02589431$ percentage points.

The 95% interval is $5.9811794 \pm 1.97338(0.02589431)$, or [5.93008%, 6.03228%] for the conditional mean unemployment rate given the two stated inputs.

### (g) A 95% prediction interval for the actual unemployment rate

The actual 2015 Q1 outcome also contains a new regression error. Under the maintained assumption that this error is uncorrelated with the estimation errors, add its variance: $se_{\text{pred}} = \sqrt{se(\widehat{U}_0)^2 + \hat{\sigma}^2} = \sqrt{0.02589431^2 + 0.2749^2} = 0.27611687$.

The prediction interval is $5.9811794 \pm 1.97338(0.27611687)$, or [5.43630%, 6.52606%].

Both intervals have the same center, but they answer different questions. Part (f) describes uncertainty about the **conditional mean**. Part (g) also allows for a **new quarterly shock**, so it is much wider. Here most of the prediction uncertainty comes from the new shock rather than estimation of the mean.

*Instructor note:* draw two intervals around 5.9812. Ask students which interval concerns an average across hypothetical repetitions with the same inputs, and which concerns the single quarter's outcome. The time-series forecast uses the maintained model; serially correlated errors would require a different prediction calculation.

## Exercise 5.14: Normality of regressors versus normality of errors

The model is $y_i = \beta_1 + \beta_2 x_i + e_i$, with independent randomly sampled pairs $(y_i, x_i)$ and an intercept. Write $S_{xx} = \sum_i (x_i - \bar{x})^2$. The useful identity is $b_2 = \beta_2 + \sum_i (x_i - \bar{x}) e_i / S_{xx}$. Assume a nondegenerate regressor, adequate sample size and the usual moment conditions needed for each expectation and limit.

### (a) Log-normal regressor, conditionally normal errors

Here $x_i$ is log-normal and $(e_i \mid x_i) \sim N(0, \sigma_e^2)$. Random sampling implies independent errors conditional on the full regressor sample $\mathbf{x}$.

**(i) Unbiasedness: yes.** Conditional mean-zero errors give $E(b_j \mid \mathbf{x}) = \beta_j$, for $j = 1, 2$. Averaging over the random regressor sample also gives $E(b_j) = \beta_j$, provided the expectations exist. The regressor does not need to have a normal distribution.

**(ii) Consistency: yes.** The log-normal regressor has finite moments and positive variance. Sample moments converge to population moments, and $\text{cov}(x_i, e_i) = 0$. Both OLS coefficients converge in probability to their true values.

**(iii) Normal conditional on x: yes.** Once the regressor values are held fixed, the coefficient estimators are linear combinations of independent normal errors. In particular, $b_2 \mid \mathbf{x} \sim N(\beta_2, \sigma_e^2 / S_{xx})$. The intercept and slope are jointly conditionally normal.

**(iv) Normal marginally: not in general.** Different random regressor samples produce different coefficient covariance matrices. Averaging their conditional normal distributions gives a mixture of normals, generally not a single normal distribution. This differs from their large-sample approximate normality.

**(v) Finite-sample $t$ tests: yes.** Conditional on any full-rank regressor sample, the usual studentized statistics have the exact $t_{N-2}$ distribution. Their conditional distribution is the same for every such sample, so the unconditional $t$ distribution is also exact. The non-normal marginal coefficient distribution in (iv) does not invalidate these tests.

**(vi) Asymptotic slope variance.** With $\mu_W = 0$ and $\sigma_W^2 = 1$, $E(x_i) = \exp(1/2)$ and $\text{var}(x_i) = [\exp(1) - 1]\exp(1) = \exp(2) - \exp(1)$. Thus the large-sample variance of $b_2$ is $\text{Avar}(b_2) = \sigma_e^2 / \{N[\exp(2) - \exp(1)]\} \approx 0.21410 \sigma_e^2 / N$. Equivalently, $\sqrt{N}(b_2 - \beta_2)$ has limiting variance $\sigma_e^2 / [\exp(2) - \exp(1)]$; the factor $1/N$ belongs to the variance of $b_2$ itself.

*Instructor note:* ask which normality assumption the exact regression $t$ result uses. It concerns the errors conditional on the regressors, not the shape of the histogram of $x$.

**(b) Normal regressor, log-normal errors**

Now $x_i \sim N(0,1)$, while the conditional error distribution is log-normal with the same parameters for every $x_i$. Its mean is $\mu_e = \exp(\mu_v + \sigma_v^2/2) > 0$ and its variance is $\sigma_e^2 = [\exp(\sigma_v^2) - 1]\mu_e^2$.

**(i) Absorb the nonzero error mean into the intercept.** Add and subtract $\mu_e$ to write $y_i = (\beta_1 + \mu_e) + \beta_2 x_i + (e_i - \mu_e) = \beta_1^* + \beta_2 x_i + e_i^*$, where $\beta_1^* = \beta_1 + \mu_e$ and $e_i^* = e_i - \mu_e$. The slope parameter is unchanged.

**(ii) Conditional moments of the centered error.** Since the conditional mean of $e_i$ is the constant $\mu_e$, $E(e_i^* \mid x_i) = E(e_i \mid x_i) - \mu_e = 0$. Subtracting a constant does not change variance, so $\text{var}(e_i^* \mid x_i) = \sigma_e^2$. The centered error is still non-normal: shifting a log-normal variable does not turn it into a normal variable.

**(iii) Is $b_2$ unbiased? Yes.** The centered model has conditional mean-zero errors. Hence $E(b_2 \mid \mathbf{x}) = \beta_2$ and $E(b_2) = \beta_2$ when these expectations exist. The fitted intercept estimates $\beta_1^*$, rather than the original $\beta_1$.

**(iv) Is $b_2$ consistent? Yes.** Random sampling, finite moments, positive regressor variance and mean-zero centered errors give $b_2 \xrightarrow{p} \beta_2$. Normal errors are not required for consistency.

**(v) Is $b_2$ conditionally normal? No, in general.** Given $\mathbf{x}$, its estimation error is a weighted sum of centered log-normal variables. It is not exactly normal in a finite sample, although a normal approximation becomes available as $N$ grows.

**(vi) Is $b_2$ marginally normal? No, in general.** A normal regressor does not make the finite-sample coefficient distribution normal when the errors are non-normal. Here the regressor weights and the denominator $S_{xx}$ are also random. The appropriate normality result is asymptotic, not exact.

**(vii) Are ordinary $t$ tests exact? No.** Without normal errors, the usual statistics are not guaranteed to follow an exact $t_{N-2}$ distribution. Under the stated random-sampling and homoskedasticity conditions, they are justified as **large-sample approximations**.

**(viii) Asymptotic slope variance.** Because $\text{var}(x_i) = 1$, the large-sample variance is $\text{Avar}(b_2) = \sigma_e^2/N$. Equivalently, the limiting variance of $\sqrt{N}(b_2 - \beta_2)$ is $\sigma_e^2$.

*Instructor note:* compare (a) and (b). In both cases, the slope is unbiased and consistent. Exact normality conditional on the regressors, and exact finite-sample $t$ inference, depend on the error distribution. Normality of the regressor does not supply them.

## Exercise 5.15: Zero correlation and conditional exogeneity

The model is $y_i = \beta_1 + \beta_2 x_i + e_i$, with independent random sampling, $x_i \sim N(0,1)$, and $E(e_i \mid x_i) = c(x_i^2 - 1)$. For a standard normal variable, $E(x_i) = 0$, $E(x_i^2) = 1$ and $E(x_i^3) = 0$.

### (a) Show that the unconditional error mean is zero

By iterated expectations, $E(e_i) = E[E(e_i \mid x_i)] = c[E(x_i^2) - 1] = c(1-1) = 0$. For $c \neq 0$, this does not make the conditional mean zero: the sign of $E(e_i \mid x_i)$ depends on whether $|x_i|$ is above or below one.

### (b) Show that the error and regressor are uncorrelated

Using the supplied covariance identity and $\mu_x = 0$, $\text{cov}(e_i, x_i) = E[x_i E(e_i \mid x_i)] = cE[x_i(x_i^2 - 1)]$. Hence $\text{cov}(e_i, x_i) = c[E(x_i^3) - E(x_i)] = 0$.

The relationship is quadratic and symmetric around zero. Its positive and negative linear contributions cancel, even though knowing $x_i$ helps predict $e_i$ when $c \neq 0$.

### (c)(i) Is the OLS slope unbiased?

**Conditional on the regressor sample: not in general, for $c \neq 0$.** The slope identity gives $E(b_2 \mid \mathbf{x}) - \beta_2 = c\sum_i (x_i - \bar{x})(x_i^2 - 1)/S_{xx}$, where $S_{xx} = \sum_i (x_i - \bar{x})^2$. This is generally nonzero, so the usual conditional unbiasedness result does not apply. For example, a regressor sample $(0, 1, 2)$ gives a conditional bias of $2c$.

**There is a special unconditional result in this exercise.** Because the joint normal regressor sample has the same distribution as its negative, the average of the conditional bias is zero: replacing every $x_i$ by $-x_i$ changes the numerator's sign but leaves $S_{xx}$ unchanged. Thus $E(b_2) = \beta_2$, provided the required expectation exists. The slope is marginally unbiased in this particular symmetric example, despite being conditionally biased.

*Instructor note:* zero error mean and zero covariance do not guarantee unbiasedness in general (Section 5.7.3). Here symmetry creates a special marginal unbiasedness result. If $c = 0$, conditional unbiasedness holds too.

### (c)(ii) Is the OLS slope consistent?

**Yes, under the usual law-of-large-numbers moment conditions.** Write $b_2 - \beta_2 = [N^{-1}\sum_i (x_i - \bar{x})e_i]/[N^{-1}\sum_i (x_i - \bar{x})^2]$. The numerator converges in probability to $\text{cov}(x_i, e_i) = 0$, while the denominator converges to $\text{var}(x_i) = 1$. Therefore $b_2 \xrightarrow{p} \beta_2$.

The conditional mean of $y_i$ is nevertheless nonlinear: $E(y_i \mid x_i) = \beta_1 + \beta_2 x_i + c(x_i^2 - 1)$. Consistency here concerns the slope of the population linear projection; it does not make this conditional mean a straight line.

## Exercise 5.16: Tuna prices, substitution and revenue

The fitted equation is $\widehat{\ln(CANS)} = -2.724 + 0.0146 RPRCE_B + 0.02649 RPRCE_C$. Sales are in thousands of cans. Relative prices are measured as percentages of brand A's price: $RPRCE_B = 100 P_B/P_A$ and $RPRCE_C = 100 P_C/P_A$. There are $N = 52$ observations, $K = 3$ coefficients and 49 residual degrees of freedom. Use $t_{0.95,49} = 1.67655$ for two-sided 10% tests. With only MR1-MR5 stated, conventional $t$ inference is approximate; exact results additionally require normal errors.

### (a) Interpret the coefficients and their expected signs

One percentage-point increase in $RPRCE_B$, holding $RPRCE_C$ fixed, increases mean log sales by $\beta_2$, corresponding to approximately $100\beta_2\%$ higher sales. The estimate is **1.46%**. For brand C, the corresponding estimate is **2.649%**. The exact changes on the exponentiated scale are $100[\exp(0.0146) - 1] = 1.4707\%$ and $100[\exp(0.02649) - 1] = 2.6844\%$.

Both signs are expected to be **positive**: when competing brands become more expensive relative to A, customers substitute toward A. A one-point change in the price ratio is not necessarily a 1% increase in the competing brand's price.

*Instructor note:* MR1-MR5 directly describe mean log sales. Interpreting these as percentage changes in mean sales also requires the retransformation factor $E[\exp(e) \mid X]$ to be constant. The normal-error corrected predictor in (f) supplies that assumption.

### (b)-(c) Is each percentage effect different from 2.5%?

Using the usual small-change approximation, test each coefficient against **0.025**, with a two-sided alternative. Calculate $t = (b_j - 0.025)/se(b_j)$ using the reported standard errors:

| Part | Null hypothesis | $t$ statistic | p-value | Decision at 10% |
|---|---|---|---|---|
| (b) | $\beta_2 = 0.025$ | -1.89781 | 0.06362 | Reject |
| (c) | $\beta_3 = 0.025$ | 0.27390 | 0.78532 | Do not reject |

The brand B effect differs from 2.5% at 10%; the brand C effect does not. If interpreting the finite 2.5% change exactly, use $\ln(1.025) = 0.0246926$ instead of 0.025. Both decisions are unchanged.

### (d) Are the two coefficients equal?

Test $H_0 : \beta_2 - \beta_3 = 0$ against $H_1 : \beta_2 - \beta_3 \neq 0$. The estimated difference is $0.0146 - 0.02649 = -0.01189$. Its standard error is $\sqrt{0.00548^2 + 0.00544^2 - 2(-0.0000143)} = 0.00939276$. Thus $t = -0.01189/0.00939276 = -1.26587$, with $p = 0.21155$. Since $|t| < 1.67655$, **do not reject equal coefficients at 10%**.

There is no contradiction with (b)-(c). A significant difference from a common benchmark for one estimate but not the other does not imply a significant difference between the two estimates. Their difference has its own standard error, including covariance.

### (e) Which brand is the closer substitute?

The **point estimates suggest brand C**: A's sales respond more strongly to a one-point rise in C's relative price. However, (d) does not establish a difference at 10%. This is suggestive evidence, not a statistically conclusive ranking.

### (f) Corrected predictions of mean sales

Let $\hat{\eta} = -2.724 + 0.0146RPRCE_B + 0.02649RPRCE_C$. The Section 4.5.3 normal-error corrected predictor is $E(\widehat{CANS} \mid X) = \exp(\hat{\eta} + \hat{\sigma}^2/2)$. Here $\hat{\sigma} = 0.5663$, so the correction added on the log scale is $0.5663^2/2 = 0.160347845$. Exponentiating only $\hat{\eta}$ would omit this correction.

For scenario (i), $\hat{\eta} = -2.724 + 0.0146(125) + 0.02649(100) = 1.7500$. Therefore predicted mean sales are $\exp(1.7500 + 0.160347845) = 6.75544$ thousand cans. Apply the same calculation to the other scenarios:

| Scenario | $RPRCE_B$ | $RPRCE_C$ | $\hat{\eta}$ | Mean sales (thousand cans) |
|---|---|---|---|---|
| (i) | 125.00 | 100.00 | 1.750000 | 6.75544 |
| (ii) | 111.11 | 88.89 | 1.252902 | 4.10929 |
| (iii) | 100.00 | 80.00 | 0.855200 | 2.76088 |

The results use the rounded relative prices specified in the book. The correction assumes conditionally normal, homoskedastic log errors; MR1-MR5 alone do not imply this particular correction.

### (g) Which of A's three prices maximizes predicted revenue?

With $P_B = 1$ dollar and $P_C = 0.80$ dollars, each candidate price of A generates the corresponding relative prices in (f). For example, if $P_A = 0.80$, then $100P_B/P_A = 125$ and $100P_C/P_A = 100$.

Revenue in dollars equals price per can times predicted sales in cans, so multiply the sales figures in (f) by 1000 first:

| A's price | Scenario | Predicted cans | Predicted weekly revenue |
|---|---|---|---|
| \$0.80 | (i) | 6755.44 | \$5404.35 |
| \$0.90 | (ii) | 4109.29 | \$3698.36 |
| \$1.00 | (iii) | 2760.88 | \$2760.88 |

**Choose 80 cents among these three candidates** to maximize predicted revenue. This compares the stated options only; it does not establish the best price over every possible price. Profit maximization would also require production costs.

### Exercise 5.17: Verify Exercise 5.1 in R

Run the blocks from top to bottom. They verify the requested results from Exercise 5.1; the PDF displays the code without executing it.

#### Enter the observations and estimate the model: part (c)

```
library(tidyverse)
options(digits = 6)
d517 <- tibble(
  y  = c(1, 1, 4, 0, 1, -2, 2),
  x2 = c(0, 1, 2, -2, 1, -2, 0),
  x3 = c(1, -2, 2, 1, -2, -1, 1)
)
model517 <- lm(y ~ x2 + x3, data = d517)
coef(model517)
```

`lm()` includes the intercept automatically. The estimates are $b_1 = 1$, $b_2 = 1$, **and** $b_3 = 0.5625$, matching 5.1(c).

#### Verify parts (e), (f), (g) and (h)

```
# (e) Square the residual standard deviation to get the variance.
summary(model517)$sigma^2
# (f) Correlation between the two explanatory variables.
cor(d517$x2, d517$x3)
# (g) Standard error: row x2, column Std. Error.
coef(summary(model517))["x2", "Std. Error"]
# (h) Residual, total and explained sums of squares, and R-squared.
SSE517 <- sum(residuals(model517)^2)
SST517 <- sum((d517$y - mean(d517$y))^2)
SSR517 <- SST517 - SSE517
c(SSE = SSE517, SST = SST517, SSR = SSR517)
summary(model517)$r.squared
```

| Requested result | R result |
|---|---|
| (e) Estimated error variance | 0.234375 |
| (f) Regressor correlation | 0 |
| (g) Standard error of $b_2$ | 0.129387 |
| (h) SSE | 0.937500 |
| (h) SST | 20.000000 |
| (h) SSR | 19.062500 |
| (h) $R^2$ | 0.953125 |

### Exercise 5.18: Food and clothing budget shares in R

#### Start up: download the data and inspect the groups

```
library(tidyverse)
options(digits = 6)

load(url(paste0(
  "https://www.principlesofeconometrics.com/poe5/",
  "data/rdata/london5.rdata"
)))
head(london5)
count(london5, nk)
```

`load()` creates `london5`, with lowercase variable names. There are **336 one-child and 514 two-child households**. Shares are fractions: 0.30 means 30%. `log()` takes natural logs.

#### (a) Expected signs

Food is expected to be a **necessity**: its expenditure share should fall as total expenditure rises, so $\beta_2 < 0$. Clothing may be a **luxury**, suggesting $\alpha_2 > 0$; a negative sign would instead classify it as a necessity.

#### (b) Estimate the two models for one-child households

```
one_child <- filter(london5, nk == 1)
food1 <- lm(wfood ~ log(totexp), data = one_child)
cloth1 <- lm(wcloth ~ log(totexp), data = one_child)
summary(food1)
summary(cloth1)
```

`filter()` keeps the requested group. In each summary, read the `log(totexp)` row for the expenditure slope and its significance.

| Outcome | Intercept | Log-expenditure slope | Slope SE | Two-sided p-value |
|---|---|---|---|---|
| Food share | 1.075486 | -0.163593 | 0.011967 | $4.28 \times 10^{-34}$ |
| Clothing share | -0.188532 | 0.064926 | 0.011844 | $8.32 \times 10^{-8}$ |

Both slopes are significant at 1%, supporting food as a necessity and clothing as a luxury. If clothing was expected to be a necessity, its positive slope might reflect purchases of better quality or discretionary clothing at larger budgets; the regression does not establish that explanation.

A 1% expenditure increase changes the fitted food share by about **-0.1636 percentage points** and clothing share by **+0.0649 percentage points**. Falling food share need not mean falling food expenditure in dollars.

### (c) A 1% lower-tail test for the food slope

Test $H_0 : \beta_2 \geq 0$ against $H_1 : \beta_2 < 0$. At the zero boundary, use the summary's `t value` and obtain its lower-tail probability with `pt()`:

```
t_food1 <- coef(summary(food1))["log(totexp)", "t value"]
p_food1 <- pt(t_food1, df = df.residual(food1))
p_food1
```

The statistic is **-13.67005** with 334 degrees of freedom and the one-sided p-value is $2.14 \times 10^{-34}$. **Reject $H_0$ at 1%**: food's share falls as expenditure rises. The alternative is the necessity claim; the null includes zero and the opposite sign.

### (d) The corresponding lower-tail test for clothing

Test $H_0 : \alpha_2 \geq 0$ against $H_1 : \alpha_2 < 0$:

```
t_cloth1 <- coef(summary(cloth1))["log(totexp)", "t value"]
p_cloth1 <- pt(t_cloth1, df = df.residual(cloth1))
p_cloth1
```

The statistic is **+5.48172**, giving a lower-tail p-value of approximately **0.99999996**. **Do not reject $H_0$**. The estimate points in the opposite direction to the alternative. Do not halve the small two-sided p-value in (b): this test asks for a negative slope, not a nonzero slope.

### (e) Fit the two-child models and compare 95% intervals

```
two_children <- filter(london5, nk == 2)
food2 <- lm(wfood ~ log(totexp), data = two_children)
cloth2 <- lm(wcloth ~ log(totexp), data = two_children)
coef(food2)
coef(cloth2)
confint(food1, "log(totexp)")
confint(food2, "log(totexp)")
confint(cloth1, "log(totexp)")
confint(cloth2, "log(totexp)")
```

The estimated slopes and their intervals are:

| Outcome | Children | Slope | 95% interval |
|---|---|---|---|
| Food | 1 | -0.163593 | [−0.187134, −0.140052] |
| Food | 2 | -0.129126 | [−0.151219, −0.107033] |
| Clothing | 1 | 0.064926 | [0.041628, 0.088224] |
| Clothing | 2 | 0.080981 | [0.058917, 0.103044] |

The food slopes suggest a weaker negative relationship for two-child households; the clothing difference looks smaller relative to its uncertainty. Both pairs of intervals overlap. **Overlap alone does not test equality**; use the interaction tests in (f).

### (f) Test whether the expenditure slopes differ by number of children

*Reading the question:* interpret the book's "95% significance level" as a **95% confidence level**, meaning tests at 5% significance.

```
# * includes log(totexp), nk, and their interaction.
food_all <- lm(wfood ~ log(totexp) * nk, data = london5)
cloth_all <- lm(wcloth ~ log(totexp) * nk, data = london5)
summary(food_all)
summary(cloth_all)

# Test the interaction against zero: the default two-sided test.
coef(summary(food_all))["log(totexp):nk", "Pr(>|t|)"]
coef(summary(cloth_all))["log(totexp):nk", "Pr(>|t|)"]
```

The expenditure slope equals the `log(totexp)` coefficient plus `nk` times the interaction coefficient. Since `nk` takes only the values 1 and 2, the interaction coefficient is exactly the difference between the two groups' slopes. Testing it against zero therefore tests equality of slopes.

| Outcome | Interaction estimate | SE | $t$ statistic | p-value |
|---|---|---|---|---|
| Food | 0.034467 | 0.016335 | 2.10999 | 0.03515 |
| Clothing | 0.016055 | 0.016252 | 0.98786 | 0.32351 |

With $850 - 4 = 846$ residual degrees of freedom, **reject equal food slopes at 5%**, but **do not reject equal clothing slopes**. The food result shows why overlapping individual confidence intervals do not imply a nonsignificant difference. These are the conventional homoskedastic regression tests.

### (g) Recover each group's slope from the interaction model

```
b_food <- coef(food_all)
b_cloth <- coef(cloth_all)

# Evaluate the slope at nk = 1 and nk = 2.
b_food["log(totexp)"] + b_food["log(totexp):nk"] * c(1, 2)
b_cloth["log(totexp)"] + b_cloth["log(totexp):nk"] * c(1, 2)
```

| Outcome | Children | Separate-regression slope | Interaction-model slope |
|---|---|---|---|
| Food | 1 | -0.163593 | -0.163593 |
| Food | 2 | -0.129126 | -0.129126 |
| Clothing | 1 | 0.064926 | 0.064926 |
| Clothing | 2 | 0.080981 | 0.080981 |

The point estimates agree because the fully interacted model allows a separate intercept and slope for each of the two groups. Its conventional standard errors pool the error variance across groups, whereas the separate regressions estimate each group's variance separately. Point-estimate agreement does not require those standard errors to be identical.

### Exercise 5.19: Food shares and expenditure elasticities in R

#### Start up and (a): when is food a necessity?

```
library(tidyverse)
library(car)  # Install once if needed: install.packages("car")
options(digits = 6)
load(url(paste0(
  "https://www.principlesofeconometrics.com/poe5/",
  "data/rdata/london5.rdata"
)))
```

The expenditure elasticity is $\varepsilon = 1 + \beta_2/w$, where $w = \beta_1 + \beta_2 \ln(TOTEXP)$ is the mean food share. For a **positive food share**, $\beta_2/w$ has the same sign as $\beta_2$. Thus $\beta_2 < 0$ if and only if $\varepsilon < 1$, which is the exercise's criterion for a necessity. A negative share would invalidate this economic argument.

#### (b) Estimate the model using all households

```
food19 <- lm(wfood ~ log(totexp), data = london5)
summary(food19)
```

The estimated intercept is **0.993839**, and the log-expenditure slope is **-0.140509**, with standard error **0.008261**. The slope's two-sided p-value is approximately $4.83 \times 10^{-56}$, and $R^2 = 0.25437$. There is strong evidence of a falling food share as total expenditure rises. The regression explains about 25.4% of sample food-share variation.

#### (c) Mean food shares at total expenditure 50, 90 and 170

```
budgets19 <- tibble(totexp = c(50, 90, 170))
predict(food19, newdata = budgets19, interval = "confidence")
```

`predict()` substitutes the three expenditure values and calculates intervals for the **mean share**, rather than for individual households. The intervals use 848 residual degrees of freedom.

| Total expenditure | Mean food share | 95% interval |
|---:|---:|---|
| 50 | 0.444165 | [0.433210, 0.455120] |
| 90 | 0.361576 | [0.355564, 0.367587] |
| 170 | 0.272214 | [0.259966, 0.284462] |

These shares correspond to approximately 44.4%, 36.2% and 27.2% of the household budget. The intervals remain in share units; multiply both endpoints by 100 to report percentages.

### (d) Elasticities and their approximate 95% intervals

The elasticity is a ratio involving two coefficients. `deltaMethod()` uses the full coefficient covariance matrix to calculate its standard error, including the covariance term. Supply the expression and give the two coefficients simple names in their model order:

```
# b1 = intercept; b2 = coefficient of log(totexp).
deltaMethod(food19, "1 + b2 / (b1 + b2 * log(50))",
            parameterNames = c("b1", "b2"))
deltaMethod(food19, "1 + b2 / (b1 + b2 * log(90))",
            parameterNames = c("b1", "b2"))
deltaMethod(food19, "1 + b2 / (b1 + b2 * log(170))",
            parameterNames = c("b1", "b2"))
```

Read `Estimate`, `SE`, `2.5 %` and `97.5 %` in each result. The function's default intervals use the normal critical value 1.96 and are **delta-method approximations**:

| Total expenditure | Elasticity | Standard error | Approximate 95% interval |
|---:|---:|---:|---|
| 50 | 0.683656 | 0.015428 | [0.653418, 0.713894] |
| 90 | 0.611399 | 0.023280 | [0.565770, 0.657028] |
| 170 | 0.483829 | 0.041074 | [0.403325, 0.564333] |

For example, at total expenditure 90, a 1% increase in total expenditure is associated with approximately **0.6114% higher food expenditure**. These are elasticities of food expenditure, not percentage changes in the food budget share.

### (e) What changes as total expenditure rises?

Both the mean food share and its expenditure elasticity fall. All three elasticity intervals lie between zero and one: food expenditure rises, but less than proportionately, and its share of the budget declines.

The mean-share interval is narrowest at 90 and wider at the two outer expenditure values. Elasticity uncertainty instead increases across these three values: its standard error rises from 0.0154 to 0.0411. Dividing by a smaller fitted food share makes the elasticity more sensitive to coefficient uncertainty.

*Instructor note:* the built-in `predict()` interval in (c) is a linear-combination interval. The elasticity interval is approximate because the transformation is nonlinear, even under normal errors.

### Exercise 5.20: A slope estimator based on two group means

Sort observations by $x$ and split the sample into equal halves. The proposed slope is the difference in group mean outcomes divided by the difference in group mean regressors. All sample sizes used below are even. The simulations illustrate the theoretical properties; simulation alone does not prove them.

#### (a) Check unbiasedness with repeated samples

```
split_slope520 <- function(x, y) {
  index <- order(x)           # Sort x and keep y paired with it.
  x <- x[index]
  y <- y[index]
  first <- seq_len(length(x) / 2)
  (mean(y[-first]) - mean(y[first])) /
    (mean(x[-first]) - mean(x[first]))
}
set.seed(520)
slopes520 <- replicate(5000, {
  x <- runif(100, 0, 10)
  y <- 2 + 3 * x + rnorm(100, sd = sqrt(1000))
  split_slope520(x, y)
})
mean(slopes520)              # Should be close to the true slope, 3.
```

`replicate()` repeats the experiment; `set.seed()` makes it reproducible. We choose intercept 2, slope 3 and error variance 1000 for this check. With this seed, the simulated mean is **2.98880**, close to the true slope 3. Conditional on the regressor sample, each group's mean error has expectation zero, so the expected numerator is $\beta_2(\bar{x}_2 - \bar{x}_1)$. Dividing by the fixed gap gives $E(\hat{\beta}_{2,mean} \mid \mathbf{x}) = \beta_2$. The estimator is therefore also marginally unbiased when its expectation exists.

#### (b)-(d) Conditional variance, marginal variance and consistency

Each half has $N/2$ independent errors. The variance of the difference in their mean errors is $4\sigma^2/N$. Writing $D = \bar{x}_2 - \bar{x}_1$, the conditional slope variance is $4\sigma^2/(ND^2)$. Its marginal variance is $(4\sigma^2/N)E(1/D^2)$, when finite, since the conditional mean is constant.

For consistency, the conditional variances must become small: a sufficient condition is $ND^2 \to \infty$ in probability. For example, if the sorted groups' mean gap approaches a nonzero constant, the slope variance tends to zero and the estimator is consistent. The following R experiment examines this condition.

### (e) Calculate both variances and the inverse-moment quantities

This helper generates one regressor sample and returns the four requested quantities. No outcomes are needed because the conditional variances depend only on the regressors and the known error variance.

```
compare520 <- function(N, sort_x = TRUE) {
  x <- runif(N, 0, 10)
  if (sort_x) x <- sort(x)
  first <- seq_len(N / 2)
  gap <- mean(x[-first]) - mean(x[first])
  sx2 <- mean((x - mean(x))^2)  # Divisor N, as requested.
  c(OLS = 1000 / (N * sx2),
    Split = 4000 / (N * gap^2),
    InvVar = 1 / sx2, InvGap = 4 / gap^2)
}
set.seed(5201)
one_sample520 <- rbind(compare520(100), compare520(500),
                       compare520(1000), compare520(5000))
data.frame(N = c(100, 500, 1000, 5000), one_sample520)
```

`OLS` and `Split` are the two conditional variances in (e)(i). `InvVar` and `InvGap` are sample realizations of the quantities whose expectations appear in (e)(ii). To estimate those expectations, average over many independent regressor samples:

```
mc520 <- rbind(
  rowMeans(replicate(1000, compare520(100))),
  rowMeans(replicate(1000, compare520(500))),
  rowMeans(replicate(1000, compare520(1000))),
  rowMeans(replicate(1000, compare520(5000)))
)
data.frame(N = c(100, 500, 1000, 5000), mc520)
```

`rowMeans()` averages each of the four returned quantities across the 1000 repetitions. In particular, mean `Split` estimates the marginal variance from (c). Monte Carlo results are approximate.

### (f) Compare efficiency and the effect of sample size

The sorted split estimator has a larger variance than OLS. OLS uses all regressor variation; the split estimator summarizes it by two group means. Both variances fall as $N$ rises. Their large-sample benchmarks are $120/N$ for OLS and $160/N$ for the split estimator:

| $N$ | Approximate OLS variance | Approximate split variance |
|---|---|---|
| 100 | 1.200 | 1.600 |
| 500 | 0.240 | 0.320 |
| 1000 | 0.120 | 0.160 |
| 5000 | 0.024 | 0.032 |

These are limiting benchmarks, not the exact simulated results. The split estimator's asymptotic variance is about one-third larger.

### (g) Explain the limiting values 0.12 and 0.16

```
# Uniform(0, 10): population variance and limiting group mean gap.
uniform_variance520 <- (10 - 0)^2 / 12
gap_limit520 <- 7.5 - 2.5
1 / uniform_variance520       # 0.12
4 / gap_limit520^2            # 0.16
```

The sample variance with divisor $N$ approaches 100/12. After sorting, the lower half approaches a uniform sample on $(0, 5)$ and the upper half approaches one on $(5, 10)$. Their means approach 2.5 and 7.5. Consequently, `InvVar` approaches **0.12** and `InvGap` approaches **0.16**. The Monte Carlo averages in (e) approach these benchmarks as sample size increases.

*Notation note:* the sample inverse-moment quantities converge in probability; their expectations are deterministic numbers. Interpreting the book's expectation notation requires the usual inverse-moment regularity conditions, under which the expectations also approach the stated constants.

### (h) What happens if observations are not sorted?

Turn sorting off in exactly the same helper:

```
set.seed(5208)
unsorted520 <- rbind(compare520(100, FALSE), compare520(500, FALSE),
                     compare520(1000, FALSE), compare520(5000, FALSE))
data.frame(N = c(100, 500, 1000, 5000), unsorted520)
```

Both unsorted halves now sample the same population, so both regressor means approach 5 and their gap approaches zero. The gap is typically of order $1/\sqrt{N}$, so $ND^2$ does not grow without bound. The denominator shrinks along with the mean-error difference: **the unsorted estimator is not consistent**. An occasional small simulated variance does not overturn this result.

Its distribution is unstable when the random mean gap is close to zero. With continuous unsorted draws, the marginal variance can be infinite because of these near-zero denominators; averaging a finite simulation does not establish a finite population variance.

*Instructor note:* sorting matters because it creates a persistent separation in regressor values. Increasing sample size without that separation does not make the difference-of-means slope reliable.

### Exercise 5.21: Rainfall, technology and wheat yield in R

#### Start up and (a): estimate the model

```
library(tidyverse)
library(car)
load(url(paste0(
  "https://www.principlesofeconometrics.com/poe5/",
  "data/rdata/toody5.rdata"
)))
model21 <- lm(y ~ trend + rain + I(rain^2) + trend:rain,
              data = toody5)
coef(summary(model21))
```

The data contain 48 annual observations, giving $48 - 5 = 43$ residual degrees of freedom. Yield is in tonnes per hectare; rainfall is in decimeters. **Use `trend`, not `t`:** trend is 0 in 1950 and rises by 0.1 per calendar year. Thus 1959 has trend 0.9 and 1995 has trend 4.5.

| Coefficient | Estimate | Standard error | $t$ statistic | p-value |
|---|---|---|---|---|
| Intercept | -2.674652 | 0.996569 | -2.68386 | 0.01029 |
| `trend` | 0.678101 | 0.191928 | 3.53309 | 0.00100 |
| `rain` | 1.508177 | 0.430729 | 3.50145 | 0.00109 |
| `I(rain^2)` | -0.157079 | 0.046622 | -3.36923 | 0.00160 |
| `trend:rain` | -0.092838 | 0.046761 | -1.98539 | 0.05350 |

#### (b) Which coefficients differ from zero?

```
p21 <- coef(summary(model21))[, "Pr(>|t|)"]
data.frame(p_value = p21, at_5_percent = p21 < 0.05,
           at_10_percent = p21 < 0.10)
```

At **5%**, all coefficients except the interaction are significant. At **10%**, all five are significant. The interaction's p-value of 0.05350 is just above 5%; retain enough digits when deciding. These are conventional two-sided regression tests.

#### (c) Interpret the signs

The positive linear rainfall term and negative squared term allow yield to rise with rainfall and then fall when rainfall becomes excessive. The positive trend coefficient captures improvement at zero rainfall; the actual trend effect is $\beta_2 + \beta_5 RAIN$ and depends on rainfall.

The **negative interaction** means yield becomes less sensitive to rainfall as time passes, consistent with developing drought-resistant varieties. It also means technological improvement is greater at lower rainfall. The negative intercept describes zero rainfall at trend zero, outside the observed rainfall range, and need not have a useful physical interpretation by itself.

### A small helper for the intervals in (d)-(e)

`deltaMethod()` calculates an estimate and its standard error from an expression. This helper uses a $t_{0.975,43}$ critical value, consistent with the chapter's regression intervals:

```
interval21 <- function(expression) {
  result <- deltaMethod(model21, expression,
                        parameterNames = c("b1", "b2", "b3", "b4", "b5"))
  estimate <- result[1, "Estimate"]
  se <- result[1, "SE"]
  critical <- qt(0.975, df = df.residual(model21))
  c(estimate = estimate, SE = se,
    lower = estimate - critical * se,
    upper = estimate + critical * se)
}
```

The names `b1` to `b5` follow the coefficient order in (a). All relevant covariances are included automatically. For a linear combination the variance calculation is exact under the usual regression assumptions; for a nonlinear ratio it is approximate.

### (d) The marginal rainfall effect in 1959 and 1995

The effect is $\beta_3 + 2\beta_4 RAIN + \beta_5 TREND$. Insert the rainfall and trend values specified for each year:

```
interval21("b3 + 2 * b4 * 2.98 + b5 * 0.9")     # 1959
interval21("b3 + 2 * b4 * 4.797 + b5 * 4.5")    # 1995
```

| Year | Rainfall | Estimated effect | SE | 95% interval |
|---|---|---|---|---|
| 1959 | 2.980 | 0.488431 | 0.145577 | [0.194847, 0.782015] |
| 1995 | 4.797 | -0.416612 | 0.154617 | [−0.728428, −0.104797] |

The units are tonnes per hectare per extra decimeter of rain. Extra rain increases predicted mean yield at the stated 1959 conditions, but decreases it at the stated 1995 conditions. Both intervals exclude zero. Both the rainfall level and the technology index differ between these observations, so their comparison does not isolate technological change alone.

### (e) Rainfall that maximizes expected yield in each year

Set the rainfall slope to zero. The maximizing rainfall is $RAIN^* = -(\beta_3 + \beta_5 TREND)/(2\beta_4)$. The fitted squared-rainfall coefficient is negative, so the stationary point is a maximum. Use the same helper:

```
interval21("-(b3 + b5 * 0.9) / (2 * b4)")      # 1959
interval21("-(b3 + b5 * 4.5) / (2 * b4)")      # 1995
```

| Year | Maximizing rainfall (dm) | SE | Approximate 95% interval |
|---|---|---|---|
| 1959 | 4.534727 | 0.235877 | [4.059036, 5.010418] |
| 1995 | 3.470879 | 0.392559 | [2.679208, 4.262549] |

The estimated yield-maximizing rainfall falls from about 4.53 dm to 3.47 dm. This is consistent with improved performance in drier conditions. The later turning point is estimated less precisely. The two individual intervals overlap; they do not constitute a formal test of the difference between the two turning points.

Actual 1959 rainfall, 2.98 dm, was below its fitted optimum; actual 1995 rainfall, 4.797 dm, was above its fitted optimum. This explains the opposite marginal effects in (d).

These are **delta-method ratio intervals**. Using a $t$ critical value does not make them exact, even under normal errors. The model describes expected yield at the stated technology level; rainfall is not a policy variable that can simply be set by a farmer.

## Exercise 5.22: Separating rainfall change and technological change

### Start up and (a): estimate the yield model

```
library(tidyverse)
library(car)
options(digits = 6)
load(url(paste0(
  "https://www.principlesofeconometrics.com/poe5/",
  "data/rdata/toody5.rdata"
)))
model22 <- lm(y ~ trend + rain + I(rain^2) + trend:rain,
              data = toody5)
coef(summary(model22))
```

This is the same specification as Exercise 5.21. The data have 48 observations and the model has 43 residual degrees of freedom. `trend` increases by **0.1 per year**, so 1960 is 1, 1974 is 2.4 and 1995 is 4.5. Rainfall is in decimeters; yield is in tonnes per hectare.

| Coefficient | Estimate | Standard error | $t$ statistic | p-value |
|---|---|---|---|---|
| Intercept | -2.674652 | 0.996569 | -2.68386 | 0.01029 |
| `trend` | 0.678101 | 0.191928 | 3.53309 | 0.00100 |
| `rain` | 1.508177 | 0.430729 | 3.50145 | 0.00109 |
| `I(rain^2)` | -0.157079 | 0.046622 | -3.36923 | 0.00160 |
| `trend:rain` | -0.092838 | 0.046761 | -1.98539 | 0.05350 |

### (b) Does extra rain increase expected yield in 1974?

At rainfall 4.576 and trend 2.4, the marginal effect is $m = \beta_3 + 2(4.576)\beta_4 + 2.4\beta_5$. Test $H_0 : m \leq 0$ against $H_1 : m > 0$:

```
rain_effect22 <- deltaMethod(model22,
  "b3 + 2 * b4 * 4.576 + b5 * 2.4",
  parameterNames = c("b1", "b2", "b3", "b4", "b5"))
t_rain22 <- rain_effect22[1, "Estimate"] / rain_effect22[1, "SE"]
rain_effect22[, c("Estimate", "SE")]
pt(t_rain22, df = df.residual(model22), lower.tail = FALSE)
```

The effect estimate is **-0.152223**, its standard error is **0.069849**, and $t = -2.17933$. The upper-tail p-value is **0.98258**, so **do not reject $H_0$ at 5%**. The estimate points toward a yield decrease, opposite to the positive effect in the alternative. Use the upper tail even though the statistic is negative. `deltaMethod()` includes all the covariances; this expression is linear in the coefficients.

### (c) Technological improvement at median rainfall

Hold rainfall at 3.8355 dm and move trend from 1 to 4.5. The yield change is $3.5(\beta_2 + 3.8355\beta_5)$. Calculate its estimate and standard error, then form a 95% interval:

```
technology22 <- deltaMethod(model22, "3.5 * (b2 + 3.8355 * b5)",
  parameterNames = c("b1", "b2", "b3", "b4", "b5"))
gain22 <- technology22[1, "Estimate"]
se_gain22 <- technology22[1, "SE"]
critical22 <- qt(0.975, df = df.residual(model22))
c(estimate = gain22, SE = se_gain22,
  lower = gain22 - critical22 * se_gain22,
  upper = gain22 + critical22 * se_gain22)
```

The estimated gain is **1.127073 tonnes per hectare**, with standard error **0.117640** and 95% interval [0.889830, 1.364316]. It is a change in the conditional mean at fixed rainfall, not a prediction interval for two actual harvests.

### (d) Is mean rainfall declining over time?

Test $H_0 : \alpha_2 \geq 0$ against $H_1 : \alpha_2 < 0$:

```
rain_model22 <- lm(rain ~ trend, data = toody5)
summary(rain_model22)
t_trend22 <- coef(summary(rain_model22))["trend", "t value"]
pt(t_trend22, df = df.residual(rain_model22))
```

The fitted equation is $\widehat{RAIN} = 4.392921 - 0.152227 TREND$. The trend standard error is **0.089075**; $t = -1.70897$ with 46 degrees of freedom. The lower-tail p-value is **0.04710**, so **reject no decline at 5%**. The two-sided summary p-value, 0.09420, answers a different question. The fitted decline is 0.152227 dm per trend unit, or **0.015223 dm per calendar year**.

### (e) Estimate mean rainfall in 1960 and 1995

```
rain_means22 <- predict(rain_model22,
                        newdata = tibble(trend = c(1, 4.5)))
rain_means22
```

The estimates are **4.240694 dm in 1960** and **3.707899 dm in 1995**, a decrease of 0.532794 dm over 35 years.

### (f) Change rainfall while holding technology at its 1960 level

Use the estimated mean rainfall from (e), keeping trend fixed at 1. The first row is the baseline; the second changes only rainfall:

```
climate_cases22 <- tibble(trend = c(1, 1), rain = rain_means22)
climate_yields22 <- predict(model22, newdata = climate_cases22)
climate_change22 <- climate_yields22[2] - climate_yields22[1]
climate_change22
```

The fitted yield change is **-0.088860 tonnes per hectare**, a decline of about 0.089 at the 1960 technology level. This is the model's rainfall-only counterfactual, not a causal estimate of the effect of climate change itself.

### (g) Change technology while holding rainfall at its 1960 mean

For the no-rainfall-change scenario, hold rainfall at the estimated 1960 mean of 4.240694 dm and move trend from 1 to 4.5:

```
technology_cases22 <- tibble(trend = c(1, 4.5),
                             rain = rep(rain_means22[1], 2))
technology_yields22 <- predict(model22, newdata = technology_cases22)
technology_change22 <- technology_yields22[2] - technology_yields22[1]
technology_change22
```

The estimated technological gain is **0.995412 tonnes per hectare**.

### (h) Compare the three changes

```
c(technology_at_median_rain = gain22,
  rainfall_change_at_1960_technology = unname(climate_change22),
  technology_at_1960_rain = unname(technology_change22))
```

The technology gains are **1.127073** in (c) and **0.995412** in (g). They use different fixed rainfall levels. With a negative interaction coefficient, the model gives a larger technology gain at the lower, median rainfall level. The rainfall-only loss in (f) is much smaller than either technological gain.

The interaction also means the two baseline changes in (f) and (g) do not simply add to the change when both inputs move:

```
combined_cases22 <- tibble(trend = c(1, 4.5), rain = rain_means22)
diff(predict(model22, newdata = combined_cases22))
```

The combined fitted gain is **1.079675**, compared with $-0.088860 + 0.995412 = 0.906552$. The additional **0.173123** is the interaction contribution $b_5 \Delta RAIN \Delta TREND$. All counterfactuals here are evaluated at estimated mean rainfall. Reading them as averages over random rainfall additionally requires the rainfall-variance contribution to be unchanged across the compared cases.

### Exercise 5.23: Quantity, quality and prices in R

#### Start up and (a): expected coefficient signs

```
library(tidyverse)
load(url(paste0(
  "https://www.principlesofeconometrics.com/poe5/",
  "data/rdata/cocaine.rdata"
)))
head(cocaine)
```

The data contain 56 historical sales observations. `price` is dollars per gram, `quant` is grams per sale, and `qual` is percentage purity. `trend` goes from 1 in 1984 to 8 in 1991.

We expect a **negative quantity coefficient** if larger purchases receive a lower unit price, and a **positive quality coefficient** if greater purity commands a premium. The time coefficient could be positive or negative: its sign depends on changes in supply, demand and other market conditions over the period.

#### (b) Estimate and interpret the model

```
model23 <- lm(price ~ quant + qual + trend, data = cocaine)
coef(summary(model23))
```

| Coefficient | Estimate | Standard error | $t$ statistic | p-value |
|---|---|---|---|---|
| Intercept | 90.846688 | 8.580254 | 10.58788 | $1.39 \times 10^{-14}$ |
| `quant` | -0.059970 | 0.010178 | -5.89194 | $2.85 \times 10^{-7}$ |
| `qual` | 0.116205 | 0.203264 | 0.57169 | 0.56999 |
| `trend` | -2.354579 | 1.386120 | -1.69868 | 0.09536 |

Holding quality and time constant, an additional gram per sale is associated with a **$0.05997 lower price per gram**. Holding quantity and time constant, one percentage point more purity is associated with a **$0.11621 higher price per gram**. The estimated quantity and quality signs match the proposed discount and quality-premium interpretations, although a positive quality estimate alone is not evidence of a statistically significant premium.

#### (c) How much price variation is explained?

```
summary(model23)$r.squared
```

The result is **0.509650**: quantity, quality and time explain about **50.97% of sample variation in price per gram around its sample mean**. This is not the percentage of sales correctly predicted.

### (d) Test for a quantity discount

Use $H_0 : \beta_2 \geq 0$ against $H_1 : \beta_2 < 0$. The alternative states the proposed unit-price discount:

```
t_quantity23 <- coef(summary(model23))["quant", "t value"]
pt(t_quantity23, df = df.residual(model23))
```

The lower-tail p-value is $1.43 \times 10^{-7}$, using $56 - 4 = 52$ degrees of freedom. **Reject at 5%**, and also at 1%. There is strong evidence of a negative conditional quantity slope. The regression supports a discount pattern, but does not establish the proposed risk mechanism as its cause.

### (e) Test for a positive quality premium

Use $H_0 : \beta_3 = 0$ against $H_1 : \beta_3 > 0$:

```
t_quality23 <- coef(summary(model23))["qual", "t value"]
pt(t_quality23, df = df.residual(model23), lower.tail = FALSE)
```

The upper-tail p-value is **0.284996**. **Do not reject at 5%**; the same decision holds at 10%. The coefficient estimate is positive, but its standard error is too large to establish a positive population premium at these levels.

### (f) The average annual price change

```
coef(model23)["trend"]
confint(model23, "trend")
```

The trend estimate implies an average **$2.35458 annual fall in price per gram**, holding quantity and quality constant. Its two-sided p-value is 0.09536: significant at 10%, but not at 5%. Possible explanations include increased supply or distribution efficiency, or weaker demand. The time trend does not distinguish these explanations; it summarizes their net association with price.

### Exercise 5.24: House size and age categories in R

#### Start up and (a): estimate the specified interaction model

```
library(tidyverse)
options(digits = 6)
load(url(paste0(
  "https://www.principlesofeconometrics.com/poe5/",
  "data/rdata/collegetown.rdata"
)))
model24 <- lm(price ~ sqft + sqft:age, data = collegetown)
coef(summary(model24))
```

The sample has 500 sales, with price in **thousands of dollars** and size in **hundreds of square feet**. `age` is a numeric category code, not the house's age in years. Use `sqft:age`, as specified in the book: `sqft * age` would also add an age main effect and estimate a different model.

| Coefficient | Estimate | Standard error | $t$ statistic | p-value |
|---|---|---|---|---|
| Intercept | -105.261319 | 13.357592 | -7.88026 | $2.08 \times 10^{-14}$ |
| `sqft` | 14.915622 | 0.651346 | 22.89969 | $9.06 \times 10^{-80}$ |
| `sqft:age` | -0.233562 | 0.073423 | -3.18103 | 0.001559 |

**All three coefficients differ from zero at 5%**, and at 1%, using the conventional two-sided tests with 497 degrees of freedom.

#### (b) Marginal effects and the category-code assumption

The size slope is $\beta_2 + \beta_3 AGE$, in thousands of dollars per extra 100 square feet. A one-category age increase changes fitted price by $\beta_3 SQFT$ thousand dollars. Treat the second expression as a discrete category change, not a derivative with respect to actual age in years.

The intercept is the fitted price at zero size; the size coefficient is the size slope at category zero. Both reference points are outside the relevant data range. The interaction says each higher age category reduces the size slope by 0.233562 thousand dollars per 100 square feet.

Using numeric codes assumes equal price effects for each category step at a fixed size, despite unequal age-band widths. Houses within a category also receive the same age adjustment.

The official collegetown definition file maps 4-5 years to category 3, 16-20 years to 6, 31-40 years to 8, and 51-75 years to 10. These codes are used below.

### (c) Size effects for houses aged 5, 20 and 40 years

Use codes 3, 6 and 8. The slope is a linear combination of two coefficients, so include their covariance when calculating its standard error:

```
b24 <- coef(model24)
V24 <- vcov(model24)
age_codes24 <- c(3, 6, 8)
size_effect24 <- b24["sqft"] + age_codes24 * b24["sqft:age"]
se_size24 <- sqrt(V24["sqft", "sqft"] +
  age_codes24^2 * V24["sqft:age", "sqft:age"] +
  2 * age_codes24 * V24["sqft", "sqft:age"])
critical24 <- qt(0.975, df = df.residual(model24))
tibble(years = c(5, 20, 40), age_code = age_codes24,
       estimate = size_effect24, SE = se_size24,
       lower = size_effect24 - critical24 * se_size24,
       upper = size_effect24 + critical24 * se_size24)
```

| Age in years | Code | Estimated size effect | 95% interval |
|---:|---:|---:|---:|
| 5 | 3 | 14.214935 | [13.206808, 15.223062] |
| 20 | 6 | 13.514247 | [12.637026, 14.391469] |
| 40 | 8 | 13.047123 | [12.145409, 13.948836] |

All effects are positive, but they decrease for older categories. For example, an extra 100 square feet for a five-year-old house corresponds to about **$14,215 higher expected price**.

### (d) Does a one-category age increase reduce price by $6000?

For each size $s$, test $H_0 : s\beta_3 = -6$ against $H_1 : s\beta_3 \neq -6$. Enter sizes 15, 30 and 45, and use **-6**, because price is in thousands of dollars:

```
sizes24 <- c(15, 30, 45)
age_change24 <- sizes24 * b24["sqft:age"]
se_age24 <- sizes24 * coef(summary(model24))["sqft:age", "Std. Error"]
t_age24 <- (age_change24 - (-6)) / se_age24
tibble(square_feet = 100 * sizes24, change = age_change24,
       t = t_age24, p = 2 * pt(-abs(t_age24), df.residual(model24)))
```

| Square feet | Estimated change ($1000) | $t$ statistic | p-value | Decision at 5% |
|---:|---:|---:|---:|---:|
| 1500 | -3.503437 | 2.26682 | 0.02383 | Reject |
| 3000 | -7.006874 | -0.45711 | 0.64779 | Do not reject |
| 4500 | -10.510310 | -1.36508 | 0.17284 | Do not reject |

For 1500-square-foot houses, the estimated decline is significantly smaller than $6000 in magnitude. For the other sizes, the uncertainty does not rule out a $6000 decline. Failure to reject does not make the estimated declines equal to $6000.

### (e) Predict the price of a 60-year-old, 2500-square-foot house

Age 60 belongs to **category 10** and 2500 square feet is **`sqft = 25`**. Request a prediction interval for an individual house, rather than a confidence interval for the conditional mean:

```
prediction24 <- predict(model24,
  newdata = tibble(sqft = 25, age = 10), interval = "prediction")
prediction24
```

The point prediction is **209.238621 thousand dollars**, with 95% prediction interval [8.772317, 409.704925] **thousand dollars**. That is approximately **$209,239**, with an interval from **$8,772 to $409,705**.

Select the observations described in the question:

```
nearby24 <- collegetown |>
  filter(age == 10, sqft >= 24.5, sqft <= 25.5) |>
  select(price, sqft, age)
nearby24
inside24 <- nearby24$price >= prediction24[1, "lwr"] &
            nearby24$price <= prediction24[1, "upr"]
inside24
sum(inside24)
```

| Floor area (square feet) | Observed price ($1000) | Inside the interval? |
|---:|---:|---|
| 2530 | 268.0 | Yes |
| 2508 | 305.0 | Yes |
| 2466 | 93.5 | Yes |
| 2539 | 150.0 | Yes |

**All four prices lie inside the interval.** However, the interval is extremely wide. Covering all four does not demonstrate precise forecasts: similar-sized houses in the same age category sell for very different prices, and many relevant characteristics are omitted. The fitted $R^2$ is about 0.6485, which does not guarantee a narrow prediction interval for an individual property.

*Data interpretation:* the book describes these as 60-year-old houses, but the file records only age category 10, covering 51-75 years. The R selection identifies that category, not exact age 60. It also compares in-sample sales with an interval at the common 2500-square-foot reference; it is a diagnostic, not independent out-of-sample validation.

### Exercise 5.25: A log-price model with a square-root term

#### Start up and (a): estimate the model

```
library(tidyverse)
library(car)
options(digits = 6)
load(url(paste0(
  "https://www.principlesofeconometrics.com/poe5/",
  "data/rdata/collegetown.rdata"
)))
model25 <- lm(log(price) ~ sqft + sqrt(sqft), data = collegetown)
coef(summary(model25))
sigma2_25 <- summary(model25)$sigma^2
sigma2_25
```

Use all 500 observations. `price` is in thousands of dollars, and `sqft` is in hundreds of square feet. The exercise assumes MR1-MR6, including conditionally normal, homoskedastic log errors. There are 497 residual degrees of freedom.

| Coefficient | Estimate | Standard error | $t$ statistic | p-value |
|---|---|---|---|---|
| Intercept | 5.045359 | 0.332218 | 15.18687 | $4.56 \times 10^{-43}$ |
| `sqft` | 0.056639 | 0.010517 | 5.38519 | $1.12 \times 10^{-7}$ |
| `sqrt(sqft)` | -0.235847 | 0.119247 | -1.97780 | 0.04850 |

**All three coefficients differ from zero at 5%** in two-sided tests. The square-root coefficient only narrowly meets that threshold. The estimated log-error variance is **0.114933**.

#### (b) The marginal effect on mean log price

At size $x = SQFT$, the slope is $m(x) = \beta_2 + \beta_3/(2\sqrt{x})$. It measures log-price change per extra 100 square feet, not a dollar-price change. A plausible expectation is a positive combined slope with diminishing gains, which would require $\beta_3 > 0$; $\beta_2$ sets the slope's large-size limit.

The fitted square-root coefficient is negative instead. Its contribution becomes less negative as size increases, so the fitted log-price slope **increases** across the sizes considered. The combined slopes are positive even though $b_3 < 0$. Do not infer a negative size effect from the square-root coefficient alone.

### (c) Log-price slopes and their 95% intervals

This helper evaluates an expression at size `x` and obtains its standard error from the full covariance matrix:

```
interval25 <- function(expression, x) {
  result <- deltaMethod(model25, expression, constants = c(x = x),
                        parameterNames = c("b1", "b2", "b3"))
  estimate <- result[1, "Estimate"]
  se <- result[1, "SE"]
  critical <- qt(0.975, df = df.residual(model25))
  c(estimate = estimate, SE = se,
    lower = estimate - critical * se,
    upper = estimate + critical * se)
}
log_effects25 <- rbind(
  interval25("b2 + b3 / (2 * sqrt(x))", 15),
  interval25("b2 + b3 / (2 * sqrt(x))", 30),
  interval25("b2 + b3 / (2 * sqrt(x))", 45))
log_effects25
```

`b1`, `b2` and `b3` follow the coefficient order in (a). The sizes 15, 30 and 45 represent 1500, 3000 and 4500 square feet. Here the expressions are linear in the coefficients, so the resulting intervals are ordinary linear-combination $t$ intervals.

| Square feet | Log-price slope | SE | 95% interval |
|---|---|---|---|
| 1500 | 0.026191 | 0.005198 | [0.015979, 0.036403] |
| 3000 | 0.035109 | 0.001555 | [0.032054, 0.038164] |
| 4500 | 0.039060 | 0.002126 | [0.034883, 0.043236] |

At 1500 square feet, an extra 100 square feet has a local effect of approximately **2.619%** on price; at 4500, about **3.906%**. These are derivative-based approximations for a finite size change.

### (d) Convert the log-price slope into a mean-price slope

Under the stated normal-error model, $E(PRICE \mid x) = \exp[\beta_1+\beta_2 x+\beta_3\sqrt{x}+\sigma^2/2]$. The chain rule multiplies this mean by the log-price slope: $\partial E(PRICE \mid x)/\partial x = m(x)E(PRICE \mid x) = S \times C$, where $S = m(x)\exp(\beta_1 + \beta_2 x + \beta_3\sqrt{x})$ and $C = \exp(\sigma^2/2)$, as defined in the question. This slope is in thousands of dollars per extra 100 square feet.

### (e)-(f) Mean-price effects and standard errors of $\widehat{S}$

Evaluate $S$ with the helper, then multiply its estimate by the log-normal correction $\widehat{C}$. For now, retain the standard error of $\widehat{S}$ as requested in (f):

```
S_expression25 <- paste0(
  "(b2 + b3/(2*sqrt(x))) * ",
  "exp(b1 + b2*x + b3*sqrt(x))")
S_results25 <- rbind(interval25(S_expression25, 15),
                     interval25(S_expression25, 30),
                     interval25(S_expression25, 45))
C25 <- exp(sigma2_25 / 2)
mean_effect25 <- C25 * S_results25[, "estimate"]
tibble(square_feet = c(1500, 3000, 4500),
       S = S_results25[, "estimate"],
       SE_S = S_results25[, "SE"], mean_price_effect = mean_effect25)
```

The correction is $\widehat{C} = 1.059150$.

| Square feet | $\widehat{S}$ | $se(\widehat{S})$ | Mean-price effect $\widehat{S}\widehat{C}$ |
|---:|---:|---:|---:|
| 1500 | 3.815848 | 0.654444 | 4.041554 |
| 3000 | 8.194208 | 0.447515 | 8.678893 |
| 4500 | 15.947805 | 1.206289 | 16.891113 |

For example, at 3000 square feet the estimated local mean-price increase is **$8679 per extra 100 square feet**. `SE_S` is not yet the standard error of the corrected effect.

### (g) Test a mean-price effect of 9 using the requested approximation

The null is a **$9000 increase per extra 100 square feet**. Following the instruction in (g), use the effects from (e) and the standard errors of $\widehat{S}$ from (f):

```
t_approx25 <- (mean_effect25 - 9) / S_results25[, "SE"]
p_approx25 <- 2 * pt(-abs(t_approx25), df = df.residual(model25))
tibble(square_feet = c(1500, 3000, 4500),
       t = t_approx25, p = p_approx25)
```

| Square feet | Approximate statistic | p-value | Decision at 5% |
|---:|---:|---:|---|
| 1500 | -7.57658 | $1.75 \times 10^{-13}$ | Reject |
| 3000 | -0.71753 | 0.47338 | Do not reject |
| 4500 | 6.54164 | $1.52 \times 10^{-10}$ | Reject |

The effects at 1500 and 4500 square feet differ from 9; the 3000-square-foot effect is compatible with 9. These tests use the exercise's simplified uncertainty calculation, refined next.

**(h) Optional: account for uncertainty in the correction factor**

Since $dC/d\sigma^2 = C/2$, the delta-method variance estimate is $\widehat{\text{var}}(\widehat{C}) = \widehat{C^2}(\hat{\sigma}^2)^2/[2(N-K)]$. Use the question's product-variance formula, including its final variance-product term:

```
var_C25 <- C25^2 * sigma2_25^2 / (2 * df.residual(model25))
var_S25 <- S_results25[, "SE"]^2
var_effect25 <- S_results25[, "estimate"]^2 * var_C25 +
  C25^2 * var_S25 + var_C25 * var_S25
se_correct25 <- sqrt(var_effect25)
p_correct25 <- 2 * pt(-abs((mean_effect25 - 9) / se_correct25),
                      df = df.residual(model25))
tibble(square_feet = c(1500, 3000, 4500),
       SE_S = S_results25[, "SE"], corrected_SE = se_correct25,
       corrected_p = p_correct25)
```

Here $\widehat{\text{var}}(\widehat{C}) = 0.000014908$.

| Square feet | SE from (f) | Corrected effect SE | Revised p-value |
|---|---|---|---|
| 1500 | 0.654444 | 0.693315 | $3.08 \times 10^{-12}$ |
| 3000 | 0.447515 | 0.475043 | 0.49939 |
| 4500 | 1.206289 | 1.279132 | $1.42 \times 10^{-9}$ |

The corrected standard errors are about 6% larger. **None of the 5% decisions changes.** The larger errors reflect both scaling the uncertainty in $S$ by $C$ and estimating the correction factor. The product and delta-method calculations are approximate, so using a $t$ reference distribution does not make these tests exact.

### Exercise 5.26: Economic conditions and presidential voting in R

#### Start up and (a): estimate the model

```
library(tidyverse)
load(url(paste0(
  "https://www.principlesofeconometrics.com/poe5/",
  "data/rdata/fair5.rdata"
)))
model26 <- lm(vote ~ growth + inflat, data = fair5)
coef(summary(model26))
```

There are 26 observations and 23 residual degrees of freedom. `vote` is the Democratic share of the two-party vote. According to the data definition, **`growth = incumb * g`** and **`inflat = incumb * p`**, where `incumb` is +1 for Democratic and -1 for Republican incumbency. Thus these regressors already contain the incumbent-party sign.

| Coefficient | Estimate | Standard error | $t$ statistic | p-value |
|---|---|---|---|---|
| Intercept | 48.757077 | 0.900200 | 54.16249 | $9.20 \times 10^{-26}$ |
| `growth` | 0.955862 | 0.165809 | 5.76484 | 0.00000715 |
| `inflat` | 0.132920 | 0.249325 | 0.53312 | 0.59906 |

At 10%, **growth differs from zero; inflation does not**. Use **two-sided tests** because the question asks whether each coefficient differs from zero, not whether it has a particular sign.

#### (b) Democratic incumbency, inflation 3%

For Democratic incumbency, the signed variables equal the economic growth and inflation rates. Predict the Democratic vote and obtain the Republican share by subtracting from 100:

```
dem_cases26 <- tibble(growth = c(-2, 0, 3), inflat = 3)
dem_vote26 <- predict(model26, newdata = dem_cases26)
tibble(economic_growth = c(-2, 0, 3),
       Democratic = dem_vote26, Republican = 100 - dem_vote26)
```

| Economic growth | Democratic vote (%) | Republican vote (%) |
|---|---|---|
| -2% | 47.2441 | 52.7559 |
| 0% | 49.1558 | 50.8442 |
| 3% | 52.0234 | 47.9766 |

Higher growth benefits the incumbent Democrats in these predictions.

### (c) Republican incumbency, inflation 3%

Reverse **both** regressor signs. The dependent variable remains the Democratic vote share:

```
rep_cases26 <- tibble(growth = -c(-2, 0, 3), inflat = -3)
dem_under_rep26 <- predict(model26, newdata = rep_cases26)
tibble(economic_growth = c(-2, 0, 3), Democratic = dem_under_rep26,
       Republican = 100 - dem_under_rep26)
```

| Economic growth | Democratic vote (%) | Republican vote (%) |
|---|---|---|
| -2% | 50.2700 | 49.7300 |
| 0% | 48.3583 | 51.6417 |
| 3% | 45.4907 | 54.5093 |

Higher growth now benefits the incumbent Republicans.

### (d) Which party does the fitted model favor?

The intercept is **48.7571 rather than 50**, giving the Republicans a fitted baseline advantage. Under the same economic conditions, the Republican incumbent's predicted share is about 2.4858 percentage points higher than the Democratic incumbent's share under Democratic incumbency. This is a point-estimate comparison; part (e) assesses its statistical uncertainty.

### (e) Compare incumbent-party shares at growth 2% and inflation 2%

```
dem_incumbent26 <- predict(model26,
  newdata = tibble(growth = 2, inflat = 2))
rep_incumbent26 <- 100 - predict(model26,
  newdata = tibble(growth = -2, inflat = -2))
c(Democratic_incumbent = unname(dem_incumbent26),
  Republican_incumbent = unname(rep_incumbent26))
```

The respective predictions are **50.93464%** and **53.42049%**. Their difference simplifies to $2\beta_1 - 100$, because the growth and inflation terms cancel. Equal shares therefore mean $H_0 : \beta_1 = 50$, against $H_1 : \beta_1 \neq 50$:

```
t_equal26 <- (coef(model26)["(Intercept)"] - 50) /
  coef(summary(model26))["(Intercept)", "Std. Error"]
2 * pt(-abs(t_equal26), df = df.residual(model26))
```

The statistic is **-1.38072** with 23 degrees of freedom and $p = 0.18064$. **Do not reject equal expected shares at 5%**. The fitted Republican advantage is not statistically established by this test. The two predictions use the same fitted model, so they must not be treated as independent estimates.

### Exercise 5.27: Painting age, size and auction prices in R

#### Start up and (a): keep sold paintings and rescale size

```
library(tidyverse)
library(car)
load(url(paste0(
  "https://www.principlesofeconometrics.com/poe5/",
  "data/rdata/ashcan_small.rdata"
)))
art27 <- ashcan_small |>
  filter(sold == 1) |>
  mutate(inchsq10 = inchsq / 10)
model27 <- lm(rhammer ~ years_old + inchsq10, data = art27)
coef(summary(model27))
```

There are **422 sold paintings**, giving 419 residual degrees of freedom in this linear-size model. `rhammer` is in thousands of dollars, while one unit of `inchsq10` is 10 square inches.

| Coefficient | Estimate | Standard error | $t$ statistic | p-value |
|---|---|---|---|---|
| Intercept | -217.853926 | 82.367388 | -2.64491 | 0.00848 |
| `years_old` | 2.755972 | 0.949332 | 2.90306 | 0.00389 |
| `inchsq10` | 1.994097 | 0.483539 | 4.12396 | 0.00004492 |

#### (b) Expected yearly price increase

```
coef(model27)["years_old"]
confint(model27, "years_old")
```

Holding size fixed, one additional year since completion is associated with **$2756 higher expected price**. The 95% interval is [0.889925, 4.622019] **thousand dollars per year**. This compares paintings of different ages; it is not a guaranteed investment return from holding a particular painting for a year.

#### (c) Expected value of an extra 10 square inches

```
coef(model27)["inchsq10"]
confint(model27, "inchsq10")
```

The estimate is **$1994**, with 95% interval [1.043633, 2.944561] **thousand dollars**, holding age fixed. The linear-size model assumes this extra value is constant at every size.

### (d) Add squared size

```
quadratic27 <- lm(rhammer ~ years_old + inchsq10 + I(inchsq10^2),
                  data = art27)
coef(summary(quadratic27))
```

| Coefficient | Estimate | Standard error | $t$ statistic | p-value |
|---|---|---|---|---|
| Intercept | -273.849704 | 84.135495 | -3.25487 | 0.00123 |
| `years_old` | 2.862085 | 0.942501 | 3.03669 | 0.00254 |
| `inchsq10` | 4.311970 | 0.959091 | 4.49589 | 0.00000898 |
| `I(inchsq10^2)` | -0.014595 | 0.005230 | -2.79085 | 0.00550 |

The squared term allows the value of additional area to change with painting size. Its negative estimate gives diminishing marginal value and eventually a fitted maximum. It differs significantly from zero at 5%. The model now has 418 residual degrees of freedom.

### (e) Does the yearly appreciation interval change much?

```
confint(model27, "years_old")
confint(quadratic27, "years_old")
```

| Model | Yearly effect ($1000) | 95% interval |
|---|---|---|
| Linear size | 2.755972 | [0.889925, 4.622019] |
| Quadratic size | 2.862085 | [1.009453, 4.714716] |

The estimate rises slightly, but the intervals are very similar. The positive age association is not materially changed by allowing curvature in painting size.

### (f) Marginal size effects at 50, 250 and 900 square inches

Use the quadratic model for the remaining parts. At rescaled size $x$, the marginal effect per 10 square inches is $\beta_3 + 2\beta_4 x$. This helper calculates its interval and also handles the nonlinear turning point in (g):

```
interval27 <- function(expression, x = 0) {
  result <- deltaMethod(quadratic27, expression, constants = c(x = x),
    parameterNames = c("b1", "b2", "b3", "b4"))
  estimate <- result[1, "Estimate"]
  se <- result[1, "SE"]
  critical <- qt(0.975, df = df.residual(quadratic27))
  c(estimate = estimate, SE = se,
    lower = estimate - critical * se,
    upper = estimate + critical * se)
}
rbind(interval27("b3 + 2*b4*x", 5),
      interval27("b3 + 2*b4*x", 25),
      interval27("b3 + 2*b4*x", 90))
```

| Area (square inches) | Marginal value ($1000) | 95% interval |
|---:|---:|---:|
| 50 | 4.166020 | [2.369059, 5.962981] |
| 250 | 3.582219 | [2.119297, 5.045141] |
| 900 | 1.684868 | [0.717177, 2.652559] |

The marginal value decreases with painting size, but all three intervals are positive. These are local slopes. For an exact finite increase of 10 square inches, the quadratic change is $b_3 + b_4(2x + 1)$, slightly below the derivative by 0.014595 thousand dollars.

### (g) Painting size that maximizes expected price

```
interval27("-10 * b3 / (2 * b4)")
```

The factor 10 converts the turning point back to square inches. The estimated maximum is at **1477.21 square inches**, with approximate 95% interval [897.69, 2056.73] **square inches**. The fitted squared-size coefficient is negative, so the turning point is a maximum. This ratio interval is a delta-method approximation even under normal regression errors.

### (h) Mean price of a 75-year-old, 100-square-inch painting

```
predict(quadratic27, newdata = tibble(years_old = 75, inchsq10 = 10),
        interval = "confidence")
```

The estimated mean price is **-17.533163 thousand dollars**, with 95% confidence interval [−61.978245, 26.911920] **thousand dollars**. This is an interval for the conditional mean, not a prediction interval for an individual auction sale.

A negative fitted price is economically implausible. It shows a limitation of this unconstrained level-price specification for this combination of age and size. Do not truncate the estimate or its interval to zero and present the result as the original regression calculation. The log-price model in Exercise 5.28 will provide a different way to model positive prices.

### (i) Age at which the fitted mean price becomes positive

Hold `inchsq10 = 10` and solve for `years_old` at fitted price zero:

```
b27 <- coef(quadratic27)
zero_age27 <- -(b27["(Intercept)"] + b27["inchsq10"] * 10 +
                b27["I(inchsq10^2)"] * 10^2) / b27["years_old"]
zero_age27
zero_age27 - 75
```

The zero-price threshold is **81.1260 years since completion**. Because the fitted age slope is positive, fitted price becomes positive above that age. For the 75-year-old painting in (h), this is approximately **6.126 additional years**, or seven whole years if restricting the answer to completed years.

This threshold is a consequence of the model's negative fitted values, not a reliable recommendation about how long to hold art. The regression describes conditional associations among sold paintings.

### Exercise 5.28: Painting prices in a log-linear model

#### Start up and (a): estimate the log-price equation

```
library(tidyverse)
library(car)
options(digits = 6)
load(url(paste0(
  "https://www.principlesofeconometrics.com/poe5/",
  "data/rdata/ashcan_small.rdata"
)))
art28 <- ashcan_small |>
  filter(sold == 1) |>
  mutate(inchsq10 = inchsq / 10)
model28 <- lm(log(rhammer) ~ years_old + inchsq10, data = art28)
coef(summary(model28))
```

The sample contains 422 sold paintings, with 419 residual degrees of freedom. Prices are in thousands of dollars and size is in tens of square inches. Percent interpretations assume that the multiplicative error factor does not vary with the regressors.

| Coefficient | Estimate | Standard error | $t$ statistic | p-value |
|---|---|---|---|---|
| Intercept | -0.245081 | 0.468667 | -0.52293 | 0.60130 |
| `years_old` | 0.021616 | 0.005402 | 4.00180 | 0.00007433 |
| `inchsq10` | 0.026725 | 0.002751 | 9.71355 | $2.99 \times 10^{-20}$ |

#### (b) Yearly percentage appreciation and its interval

```
100 * coef(model28)["years_old"]
100 * confint(model28, "years_old")
100 * expm1(coef(model28)["years_old"])
100 * expm1(confint(model28, "years_old"))
```

The usual approximation is **2.1616% per year**, with 95% interval [1.0999%, 3.2234%]. `expm1(x)` calculates $\exp(x) - 1$. Using the exact one-year transformation gives **2.1852%**, with transformed interval [1.1059%, 3.2759%]. This is an age association holding size constant, not a guaranteed return on a particular painting.

#### (c) Does extra size increase value by more than 2%?

Using the usual small-change approximation, test $H_0 : \beta_3 \leq 0.02$ against $H_1 : \beta_3 > 0.02$:

```
t_size28 <- (coef(model28)["inchsq10"] - 0.02) /
  coef(summary(model28))["inchsq10", "Std. Error"]
pt(t_size28, df = df.residual(model28), lower.tail = FALSE)
```

The upper-tail statistic is **2.44431**, with $p = 0.007462$. **Reject at 5%**: the fitted effect exceeds 2% per extra 10 square inches. For an exact finite 2% change, use $\ln(1.02)$ rather than 0.02 as the null boundary; the decision is unchanged.

### (d) Add squared size

```
quadratic28 <- lm(log(rhammer) ~ years_old + inchsq10 + I(inchsq10^2),
                  data = art28)
coef(summary(quadratic28))
```

| Coefficient | Estimate | Standard error | $t$ statistic | p-value |
|---|---|---|---|---|
| Intercept | -0.957484 | 0.460544 | -2.07903 | 0.03822 |
| `years_old` | 0.022966 | 0.005159 | 4.45163 | 0.00001095 |
| `inchsq10` | 0.056214 | 0.005250 | 10.70764 | $8.31 \times 10^{-24}$ |
| `I(inchsq10^2)` | -0.000186 | 0.000029 | -6.48657 | $2.48 \times 10^{-10}$ |

The significant negative squared term allows diminishing percentage gains from extra area. Residual degrees of freedom: 418.

### (e) Compare the yearly appreciation intervals

```
100 * confint(model28, "years_old")
100 * confint(quadratic28, "years_old")
```

The approximate yearly interval changes from [1.0999%, 3.2234%] to [1.2825%, 3.3107%]. The estimate rises from 2.1616% to 2.2966%; the positive age association changes little.

### (f) Repeat the size test at three painting sizes

At rescaled area $x$, the log-price slope is $m(x) = \beta_3 + 2\beta_4 x$. Test $H_0 : m(x) \leq 0.02$ against $H_1 : m(x) > 0.02$. One helper repeats the same calculation:

```
size_test28 <- function(x) {
  result <- deltaMethod(quadratic28, "b3 + 2*b4*x",
    parameterNames = c("b1", "b2", "b3", "b4"), constants = c(x = x))
  t <- (result[1, "Estimate"] - 0.02) / result[1, "SE"]
  c(effect = result[1, "Estimate"], SE = result[1, "SE"], t = t,
    p = pt(t, df = df.residual(quadratic28), lower.tail = FALSE))
}
rbind(size_test28(5), size_test28(25), size_test28(90))
```

| Area (square inches) | Log-price slope | $t$ statistic | Upper-tail p-value |
|---|---|---|---|
| 50 | 0.054357 | 6.86586 | $1.20 \times 10^{-11}$ |
| 250 | 0.046930 | 6.61041 | $5.86 \times 10^{-11}$ |
| 900 | 0.022791 | 1.03568 | 0.15048 |

**Reject at 50 and 250 square inches; do not reject at 900.** The fitted effect falls with size; at 900, uncertainty prevents rejection. Using the exact finite-change expression $\beta_3 + (2x+1)\beta_4$ and null boundary $\ln(1.02)$ gives the same decisions.

### (g) Size that maximizes fitted price

```
turning28 <- deltaMethod(quadratic28, "-10*b3/(2*b4)",
  parameterNames = c("b1", "b2", "b3", "b4"))
critical28 <- qt(0.975, df = df.residual(quadratic28))
turn_est28 <- turning28[1, "Estimate"]
turn_se28 <- turning28[1, "SE"]
c(estimate = turn_est28, SE = turn_se28,
  lower = turn_est28 - critical28 * turn_se28,
  upper = turn_est28 + critical28 * turn_se28)
```

The maximizing area is **1513.70 square inches**, with approximate 95% interval [1255.11, 1772.29] **square inches**. The factor 10 reverses the size rescaling. Since exponentiation is increasing, the fitted log-price maximum also maximizes its exponentiated value. With constant log-error variance, the log-normal mean-price correction leaves the maximizing size unchanged. The ratio interval is a delta-method approximation.

### (h) Exponentiated fitted log price and its uncertainty

Follow the estimator explicitly requested in the question: $\widehat{P} = \exp(E[\ln(\widehat{RHAMMER}) \mid X])$. Its delta-method standard error is the estimate times the standard error of the fitted log mean:

```
log_prediction28 <- predict(quadratic28,
  newdata = tibble(years_old = 75, inchsq10 = 10), se.fit = TRUE)
price_est28 <- exp(log_prediction28$fit)
price_se28 <- price_est28 * log_prediction28$se.fit
c(estimate = unname(price_est28), SE = unname(price_se28),
  lower = unname(price_est28 - critical28 * price_se28),
  upper = unname(price_est28 + critical28 * price_se28))
```

The point estimate is **3.700872 thousand dollars**, its standard error is **0.458049**, and the approximate symmetric 95% interval is [2.800505, 4.601240] **thousand dollars**.

**Interpretation of the requested estimator:** this is the conditional geometric mean, and the conditional median under normal log errors. The arithmetic conditional mean includes the additional factor $\exp(\sigma^2/2)$; its corrected point estimate here would be 15.765699 thousand dollars. These quantities differ substantially because estimated log-error variance is large.

An alternative positive interval for the requested exponentiated log mean is obtained by transforming the log-scale confidence interval:

```
exp(predict(quadratic28,
  newdata = tibble(years_old = 75, inchsq10 = 10),
  interval = "confidence"))
```

This gives [2.901661, 4.720212] **thousand dollars**. It transforms the endpoints instead of approximating directly on the price scale. Neither interval is an individual-sale prediction interval.

## Exercise 5.29: Crime and punishment in the 1986 county data

### Download, select the year and estimate the requested model

```
library(tidyverse)
load(url(paste0(
  "https://www.principlesofeconometrics.com/poe5/",
  "data/rdata/crime.rdata"
)))
crime86 <- filter(crime, year == 86)
nrow(crime86)
model29 <- lm(lcrmrte ~ prbarr + prbconv + prbpris + polpc + wcon,
              data = crime86)
coef(summary(model29))
summary(model29)$r.squared
```

The year is coded **86**, not 1986. The selected sample contains 90 counties; the model has 84 residual degrees of freedom. Use the regressors in levels as requested, even though the file also contains logged versions of several of them.

### Expected signs and regression results

Arrest, conviction and prison-sentence probabilities would usually be expected to deter crime, suggesting negative coefficients. More police might also deter crime, and better construction wages could raise the opportunity cost of illegal activity, suggesting negative signs for those variables as well. The latter two may also reflect other county characteristics and reverse causality.

| Coefficient | Estimate | Standard error | $t$ statistic | p-value |
|---|---|---|---|---|
| Intercept | -3.363608 | 0.317195 | -10.60422 | $3.62 \times 10^{-17}$ |
| `prbarr` | -2.043717 | 0.265267 | -7.70438 | $2.36 \times 10^{-11}$ |
| `prbconv` | -0.993153 | 0.090752 | -10.94365 | $7.71 \times 10^{-18}$ |
| `prbpris` | 0.300371 | 0.402710 | 0.74587 | 0.45783 |
| `polpc` | 150.230941 | 13.534264 | 11.10004 | $3.79 \times 10^{-18}$ |
| `wcon` | 0.002114 | 0.000761 | 2.77853 | 0.00674 |

At 5%, all slopes except `prbpris` differ from zero in conventional two-sided tests. Arrest and conviction have the expected negative signs. Prison sentencing, police and construction wages have positive estimates; only the latter two are significant. The model explains **73.74%** of sample variation in log crime rates.

### Interpret meaningful changes rather than compare coefficient sizes

The dependent variable is logged. For a regressor change $\Delta x$, $100[\exp(b\Delta x) - 1]$ gives the fitted percentage change on the exponentiated scale:

```
# Arrest, conviction and prison probabilities: +0.10 each.
# Police per capita: +0.001. Weekly construction wages: +100 dollars.
changes29 <- c(0.10, 0.10, 0.10, 0.001, 100)
100 * expm1(coef(model29)[-1] * changes29)
```

| Change, holding other regressors fixed | Fitted percentage change |
|---|---|
| Arrest probability +0.10 | -18.48% |
| Conviction probability +0.10 | -9.45% |
| Prison-sentence probability +0.10 | +3.05% |
| Police per capita +0.001 | +16.21% |
| Weekly construction wage +100 dollars | +23.54% |

The percentage interpretation for the arithmetic mean additionally requires a constant retransformation factor. Raw coefficient magnitudes cannot rank importance because the measurement scales differ.

### Report: deterrence evidence and the positive police coefficient

**Arrest and conviction probabilities provide the clearest negative associations with crime in this specification.** Both have very small p-values and sizeable fitted effects for a 10-percentage-point change. There is no comparable evidence for the prison-sentence probability: its estimate is positive and imprecise.

The positive police coefficient does not establish that police cause crime. Counties with more crime may hire more police; urbanization and other omitted characteristics may increase both police staffing and recorded crime. Those mechanisms can produce a positive OLS association even if policing has a deterrent causal effect.

The positive wage coefficient also conflicts with the simplest opportunity-cost prediction. Construction wages may proxy for urban economic conditions, prices or omitted opportunities for property crime. This regression does not distinguish those explanations.

These are **cross-county conditional associations**, not identified causal effects of changing punishment or staffing. The arrest and conviction results are the strongest deterrence-consistent findings here, but policy effects would need stronger identification and attention to the error assumptions. A high $R^2$ does not resolve those interpretation issues.

### Exercise 5.30: Could each advertising level be optimal for Andy?

#### Start up: estimate the quadratic advertising model

```
library(tidyverse)
library(car)
load(url(paste0(
  "https://www.principlesofeconometrics.com/poe5/",
  "data/rdata/andy.rdata"
)))
model30 <- lm(sales ~ price + advert + I(advert^2), data = andy)
coef(model30)
```

The fitted equation is $\widehat{SALES} = 109.719036 - 7.640000PRICE + 12.151236ADVERT - 2.767963ADVERT^2$. There are 75 observations and 71 residual degrees of freedom. Both sales and advertising are measured in thousands of dollars.

The book's optimality condition is $\beta_3 + 2\beta_4 ADVERT_0 = 1$: marginal revenue equals one advertising dollar per dollar spent. This maximizes expected sales revenue minus advertising expenditure at fixed price in the stated model; production costs are not included.

#### (a)-(c) Test the three candidate levels

For each proposed $a$, test $H_0 : \beta_3 + 2a\beta_4 = 1$ against $H_1 : \beta_3 + 2a\beta_4 \neq 1$. The expression is a linear combination of coefficients, so its standard error must include their covariance:

```
optimality30 <- function(a) {
  result <- deltaMethod(model30, "b3 + 2*b4*a",
    parameterNames = c("b1", "b2", "b3", "b4"), constants = c(a = a))
  marginal_revenue <- result[1, "Estimate"]
  se <- result[1, "SE"]
  t <- (marginal_revenue - 1) / se
  c(advert = a, marginal_revenue = marginal_revenue, SE = se, t = t,
    p = 2 * pt(-abs(t), df = df.residual(model30)))
}
rbind(optimality30(1.75), optimality30(1.9), optimality30(2.3))
```

| Advertising ($1000) | Marginal revenue | SE | $t$ statistic | p-value |
|---:|---:|---:|---:|---:|
| 1.75 | 2.463365 | 0.680851 | 2.14932 | 0.03502 |
| 1.90 | 1.632976 | 0.654190 | 0.96757 | 0.33654 |
| 2.30 | -0.581395 | 1.054348 | -1.49988 | 0.13808 |

At 5%, **reject optimality at 1.75** thousand dollars. **Do not reject at 1.90 or 2.30** thousand dollars. At 1.75, estimated marginal revenue significantly exceeds one, suggesting that increasing advertising would improve the book's objective. At 2.30, the point estimate is negative, but its uncertainty is large enough that the test still does not rule out a true value of one.

Failure to reject does not prove that either candidate is optimal, and it does not imply that two different values maximize the same population objective. The tests concern compatibility with the data at their respective null values. The negative fitted squared-advertising coefficient gives the concavity needed for the fitted stationary point to be a maximum.

### Exercise 5.31: Can Bill reach the university on time?

#### (a) Download the data and estimate the model

```
library(tidyverse)
library(car)
load(url(paste0(
  "https://www.principlesofeconometrics.com/poe5/",
  "data/rdata/commute5.rdata"
)))
model31 <- lm(time ~ depart + reds + trains, data = commute5)
coef(summary(model31))
```

There are **249 working days** and **245 residual degrees of freedom**. `time` is minutes of travel; `depart` is minutes after 6:30 AM. The fitted equation is $\widehat{TIME} = 20.870136 + 0.368125DEPART + 1.521943REDS + 3.023696TRAINS$.

| Coefficient | Estimate | Standard error | p-value |
|---|---|---|---|
| Intercept | 20.870136 | 1.675760 | $6.55 \times 10^{-28}$ |
| `depart` | 0.368125 | 0.035102 | $1.70 \times 10^{-21}$ |
| `reds` | 1.521943 | 0.185038 | $1.15 \times 10^{-14}$ |
| `trains` | 3.023696 | 0.634012 | 0.00000318 |

Holding the other regressors fixed, departing one minute later is associated with **0.3681 extra minutes of travel**. Each additional red light adds **1.5219 minutes** and each train **3.0237 minutes** to fitted travel time. The intercept predicts **20.8701 minutes** for a 6:30 AM departure with no red lights and no trains; it is not the average journey across all departure times.

#### (b) Confidence intervals and precision

```
confint(model31)
```

| Parameter | Lower 95% endpoint | Upper 95% endpoint |
|---|---|---|
| $\beta_1$ | 17.569402 | 24.170871 |
| $\beta_2$: departure | 0.298985 | 0.437265 |
| $\beta_3$: red lights | 1.157475 | 1.886411 |
| $\beta_4$: trains | 1.774887 | 4.272505 |

All intervals exclude zero. Departure time is estimated relatively precisely. The train effect is less precise relative to its estimate: its plausible range spans roughly 1.77 to 4.27 minutes per train. Whether that is precise enough depends on Bill's scheduling needs; statistical significance alone does not answer that question.

### (c)-(f) Tests about delays

Use the same calculation for each coefficient or linear combination. `deltaMethod()` includes the coefficient covariances in the standard error. Here every expression is linear, so its variance calculation is exact under the usual regression assumptions.

```
test31 <- function(expression, null = 0, alternative = "two.sided") {
  result <- deltaMethod(model31, expression,
    parameterNames = c("b1", "b2", "b3", "b4"))
  estimate <- result[1, "Estimate"]
  se <- result[1, "SE"]
  t <- (estimate - null) / se
  df <- df.residual(model31)
  p <- switch(alternative,
    less = pt(t, df), greater = pt(t, df, lower.tail = FALSE),
    two.sided = 2 * pt(-abs(t), df))
  c(estimate = estimate, SE = se, t = t, p = p)
}
test31("b3", 2, "less")
test31("b4", 3)
test31("30*b2", 10, "less")
test31("b4 - 3*b3", 0, "less")
```

The parameter order is intercept, departure, red lights and trains. `less` selects a lower-tail test; the default is a two-sided test. For an inequality null, calculate the reference distribution at its boundary, where equality holds.

| Part | Null and alternative | $t$ statistic | p-value | Decision |
|---|---|---|---|---|
| (c) | $H_0 : \beta_3 \geq 2$; $H_1 : \beta_3 < 2$ | -2.58356 | 0.00518 | Reject at 5% |
| (d) | $H_0 : \beta_4 = 3$; $H_1 : \beta_4 \neq 3$ | 0.03737 | 0.97022 | Do not reject at 10% |
| (e) | $H_0 : 30\beta_2 \geq 10$; $H_1 : 30\beta_2 < 10$ | 0.99116 | 0.83871 | Do not reject at 5% |
| (f) | $H_0 : \beta_4 - 3\beta_3 \geq 0$; $H_1 : \beta_4 - 3\beta_3 < 0$ | -1.82503 | 0.03461 | Reject at 5% |

**(c)** There is evidence that the expected delay per red light is less than two minutes.

**(d)** The train estimate is very close to three minutes. The data are compatible with a three-minute delay; this does not prove equality.

**(e)** Moving departure from 7:00 to 7:30 AM changes `depart` from 30 to 60. Fitted travel time increases by **11.0438 minutes**, holding red lights and trains fixed. There is no evidence that the expected increase is less than ten minutes.

**(f)** The estimated difference between a train delay and three red-light delays is **-1.5421 minutes**. There is evidence that the expected train delay is less than three times the red-light delay.

### (g) Six red lights, one train and a 7:45 AM meeting

A 7:00 AM departure leaves **45 minutes** for travel. Define $m = \beta_1 + 30\beta_2 + 6\beta_3 + \beta_4$ and test $H_0 : m \leq 45$ against $H_1 : m > 45$:

```
test31("b1 + 30*b2 + 6*b3 + b4", 45, "greater")
```

Estimated expected travel time is **44.0692 minutes**, with standard error **0.53927**. The statistic is **-1.72596** and the upper-tail p-value is **0.95719**. **Do not reject at 5%**. There is no evidence, in this test, that expected travel time exceeds 45 minutes. Expected arrival is approximately 7:44 AM. Failure to reject the null does not establish that Bill will be on time.

### (h) Put the claim requiring evidence in the alternative

If lateness is costly, treating an acceptable expected travel time as the default may be a poor decision rule. Reverse the hypotheses: $H_0 : m \geq 45$ against $H_1 : m < 45$. Bill then requires evidence that expected travel time is below the available 45 minutes.

```
test31("b1 + 30*b2 + 6*b3 + b4", 45, "less")
```

The statistic remains **-1.72596**, but the lower-tail p-value is **0.04281**. **Reject at 5%**: expected travel time is below 45 minutes. Both tests can have these outcomes because their null hypotheses and rejection regions differ.

**Expected time is not an individual journey.** A hypothesis test about $E(TIME \mid X)$ does not provide a guarantee against lateness. For comparison, obtain an individual-journey prediction interval:

```
predict(model31,
  newdata = tibble(depart = 30, reds = 6, trains = 1),
  interval = "prediction")
```

The two-sided 95% prediction interval is [31.6177, 56.5208] **minutes**, which extends well beyond 45 minutes. Bill needs a larger safety margin if being late is unacceptable. This prediction also assumes the stated numbers of red lights and trains; uncertainty about those counts would add another scheduling problem.

### Exercise 5.32: Isolate the departure-time coefficient

#### (a) Remove the part of departure time associated with lights and trains

Use the `commute5` data loaded in Exercise 5.31. Each auxiliary regression includes an intercept.

```
depart32 <- lm(depart ~ reds + trains, data = commute5)
coef(summary(depart32))
depart_res32 <- resid(depart32)
```

| Coefficient | Estimate | Standard error | p-value |
|---|---|---|---|
| Intercept | 21.181412 | 2.727789 | $2.20 \times 10^{-13}$ |
| `reds` | -0.369905 | 0.335267 | 0.27097 |
| `trains` | 8.455609 | 1.017610 | $6.53 \times 10^{-15}$ |

At 5%, the **train coefficient** differs from zero; the red-light coefficient does not. The intercept is also significant. The train coefficient is an association: departures one train apart are associated with an 8.4556-minute difference in departure time, holding red lights fixed. It does not show that a train encountered during the journey causes Bill to leave home later. Departure time may affect train encounters, and both may reflect the timetable or traffic patterns.

#### (b) Remove the part of travel time associated with lights and trains

```
time32 <- lm(time ~ reds + trains, data = commute5)
coef(summary(time32))
time_res32 <- resid(time32)
```

| Coefficient | Without departure | With departure (5.31) |
|---|---|---|
| Intercept | 28.667545 | 20.870136 |
| `reds` | 1.385772 | 1.521943 |
| `trains` | 6.136418 | 3.023696 |

The red-light estimate changes modestly. The train estimate more than doubles when departure time is omitted. Later departures are associated with both longer travel and more train encounters, so the reduced model assigns part of the departure-time association to trains. Numerically, the omitted-variable identities give $1.521943 + 0.368125(-0.369905) = 1.385772$ for lights and $3.023696 + 0.368125(8.455609) = 6.136418$ for trains.

### (c) Regress the two residuals and correct the standard error

```
partial32 <- lm(time_res32 ~ depart_res32 - 1)
coef(summary(partial32))

# Four parameters were estimated in the original complete model.
df32 <- nrow(commute5) - 4
se32 <- sqrt(sum(resid(partial32)^2) /
  (df32 * sum(depart_res32^2)))
coef(partial32)
se32
```

The residual-on-residual slope is **0.368125**, exactly the departure-time slope from the full model. This is the **Frisch-Waugh-Lovell theorem**: remove the linear associations with the other regressors from both variables, then regress what remains. The residual regression has no intercept because both sets of residuals already have sample mean zero.

The raw residual-regression output reports standard error **0.034889**, using $249 - 1 = 248$ residual degrees of freedom. The correct calculation uses **245**, accounting for all four parameters in the full regression. Corrected standard error: **0.035102**, equal to the full-model value.

Verify the two results directly:

```
full32 <- lm(time ~ depart + reds + trains, data = commute5)
c(partial_slope = unname(coef(partial32)),
  full_slope = coef(full32)["depart"])
c(corrected_SE = se32,
  full_SE = coef(summary(full32))["depart", "Std. Error"])
```

**Instructor prompt:** what is left in `depart_res32`? It is the part of observed departure time that cannot be predicted by a linear regression on red lights and trains. We compare that with the corresponding unexplained part of travel time. This gives the meaning of controlling for the other regressors in OLS; it does not by itself establish a causal departure-time effect.

### Exercise 5.33: Education, experience and log wages

#### (a) Estimate the model and inspect significance

```
library(tidyverse)
library(car)
load(url(paste0(
  "https://www.principlesofeconometrics.com/poe5/",
  "data/rdata/cps5_small.rdata"
)))
model33 <- lm(log(wage) ~ educ + I(educ^2) + exper + I(exper^2) +
                I(educ * exper), data = cps5_small)
coef(summary(model33))
b33 <- coef(model33)
```

There are **1200 observations**, six coefficients and **1194 residual degrees of freedom**. Parameter order is intercept, education, education squared, experience, experience squared and interaction.

| Parameter | Estimate | Standard error | p-value |
|---|---|---|---|
| $\beta_1$ | 1.037933 | 0.275741 | 0.000175 |
| $\beta_2$: `educ` | 0.089539 | 0.031082 | 0.004038 |
| $\beta_3$: `I(educ^2)` | 0.001458 | 0.000924 | 0.114855 |
| $\beta_4$: `exper` | 0.044879 | 0.007297 | $1.06 \times 10^{-9}$ |
| $\beta_5$: `I(exper^2)` | -0.000468 | 0.000076 | $1.01 \times 10^{-9}$ |
| $\beta_6$: `I(educ * exper)` | -0.001010 | 0.000379 | 0.007803 |

At the conventional 10%, 5% and 1% levels, **all coefficients except education squared differ from zero**. Its p-value is about 11.49%, so it is not significant even at 10%. These are individual coefficient tests, not a joint test of all education terms.

#### (b) Marginal effect of education

The log-wage derivative is $ME_{EDUC} = \beta_2 + 2\beta_3 EDUC + \beta_6 EXPER$. Its fitted value is $0.089539 + 0.002917 EDUC - 0.001010 EXPER$. The fitted education effect increases with education and decreases with experience. Multiplying by 100 gives an approximate percentage effect per extra year of education, holding experience fixed. The derivative describes a local change; an exact finite one-year change also includes the extra squared-term contribution $\beta_3$.

### (c) Distribution of education effects across the sample

```
effects33 <- cps5_small |>
  mutate(educ_effect = b33[2] + 2 * b33[3] * educ + b33[6] * exper,
         exper_effect = b33[4] + 2 * b33[5] * exper + b33[6] * educ)
ggplot(effects33, aes(x = 100 * educ_effect)) +
  geom_histogram(bins = 25, color = "white") +
  labs(x = "Education effect (approximate percent per year)",
       y = "Number of workers")
quantile(effects33$educ_effect, c(0.05, 0.50, 0.95))
range(effects33$educ_effect)
sum(effects33$educ_effect < 0)
```

| Education effect | 5th percentile | Median | 95th percentile |
|---|---|---|---|
| Log-wage derivative | 0.080082 | 0.108431 | 0.133619 |
| Approximate percent per year | 8.0082% | 10.8431% | 13.3619% |

The histogram shows heterogeneity across observed education and experience combinations. **All 1200 fitted education effects are positive**, ranging from 0.035654 to 0.147871. The percentile range describes variation in fitted effects across workers; it is not a confidence interval for a single parameter.

### (d)-(e) Experience effects and their distribution

The derivative is $ME_{EXPER} = \beta_4 + 2\beta_5 EXPER + \beta_6 EDUC$, estimated as $0.044879 - 0.000936 EXPER - 0.001010 EDUC$. It decreases with both experience and education in this fitted model.

```
ggplot(effects33, aes(x = 100 * exper_effect)) +
  geom_histogram(bins = 25, color = "white") +
  labs(x = "Experience effect (approximate percent per year)",
       y = "Number of workers")
quantile(effects33$exper_effect, c(0.05, 0.50, 0.95))
sum(effects33$exper_effect < 0)
```

| Experience effect | 5th percentile | Median | 95th percentile |
|---|---|---|---|
| Log-wage derivative | -0.010376 | 0.008419 | 0.027931 |
| Approximate percent per year | -1.0376% | 0.8419% | 2.7931% |

The fitted effects span both signs: **340 workers (28.33%)** have a negative experience derivative. The quadratic specification allows fitted log wages to decline after an experience threshold. This cross-sectional association is not proof that the same workers' wages would fall if they gained another year of experience.

### (f) Compare David and Svetlana now

David has $(EDUC, EXPER) = (17, 8)$ and Svetlana $(16, 18)$. Let $D$ be David's expected log wage minus Svetlana's: $D = \beta_2 + 33\beta_3 - 10\beta_4 - 260\beta_5 - 152\beta_6$. Test $H_0 : D \leq 0$ **against** $H_1 : D > 0$ at 5%.

The helper below tests linear combinations with their correct standard errors. `greater` selects an upper-tail test.

```
test33 <- function(expression, alternative = "two.sided") {
  result <- deltaMethod(model33, expression,
    parameterNames = paste0("b", 1:6))
  estimate <- result[1, "Estimate"]
  se <- result[1, "SE"]
  t <- estimate / se
  df <- df.residual(model33)
  p <- if (alternative == "greater") {
    pt(t, df, lower.tail = FALSE)
  } else {
    2 * pt(-abs(t), df)
  }
  c(estimate = estimate, SE = se, t = t, p = p)
}
test33("b2 + 33*b3 - 10*b4 - 260*b5 - 152*b6", "greater")
```

The estimated difference is **-0.035885**, standard error **0.021489**, statistic **-1.66990** and p-value **0.95240**. **Do not reject**: there is no evidence that David has the higher expected log wage now. The point estimate favors Svetlana.

### (g) Compare them after eight additional years of experience

The new pairs are $(17, 16)$ and $(16, 26)$. Education is unchanged. The new difference is $D_8 = \beta_2 + 33\beta_3 - 10\beta_4 - 420\beta_5 - 144\beta_6$. Again test $H_0 : D_8 \leq 0$ against $H_1 : D_8 > 0$:

```
test33("b2 + 33*b3 - 10*b4 - 420*b5 - 144*b6", "greater")
```

The estimated difference is **0.030917**, standard error **0.014991**, statistic **2.06237** and p-value **0.01969**. **Reject at 5%**: David now has the higher fitted expected log wage. The difference changes by $-160\beta_5 + 8\beta_6$. Equal additions to experience do not cancel in its square or its interaction with different education levels. The negative squared-experience term makes Svetlana's extra experience less valuable at these higher levels. This is a comparison within the fitted model, assuming its coefficients continue to apply eight years later.

### (h) Do Wendy and Jill have the same marginal experience effect?

Wendy has $(EDUC, EXPER) = (12, 17)$ and Jill $(16, 11)$. Subtract Jill's experience derivative from Wendy's: $ME_W - ME_J = 12\beta_5 - 4\beta_6$. Test $H_0 : 12\beta_5 - 4\beta_6 = 0$ against $H_1 : 12\beta_5 - 4\beta_6 \neq 0$:

```
test33("12*b5 - 4*b6")
```

The difference is **-0.001575**, standard error **0.001533**, statistic **-1.02730** and two-sided p-value **0.30449**. **Do not reject at 5%**. The fitted effects differ by about -0.1575 percentage points per year, but this is not statistically distinguishable from zero. This tests derivatives, not log-wage levels.

### (i) When does Jill's marginal experience effect become negative?

Hold Jill's education at 16 and solve $\beta_4 + 2\beta_5 EXPER + 16\beta_6 = 0$. The experience threshold is $EXPER^* = -(\beta_4 + 16\beta_6)/(2\beta_5)$. She already has 11 years, so the additional years required are $A = -(\beta_4 + 16\beta_6)/(2\beta_5) - 11$.

```
years33 <- deltaMethod(model33, "-(b4 + 16*b6)/(2*b5) - 11",
  parameterNames = paste0("b", 1:6))
estimate33 <- years33[1, "Estimate"]
se33 <- years33[1, "SE"]
critical33 <- qt(0.975, df = df.residual(model33))
c(additional_years = estimate33, SE = se33,
  lower = estimate33 - critical33 * se33,
  upper = estimate33 + critical33 * se33)
```

The threshold is **30.6771 total years of experience**, or **19.6771 additional years** for Jill. Because the fitted squared-experience coefficient is negative, her marginal effect is zero at the threshold and negative beyond it.

The delta-method standard error for additional years is **1.895713**; the approximate 95% interval is [15.9578, 23.3964] **additional years**. Adding 11 gives [26.9578, 34.3964] **total years**. This is an interval for the fitted turning point, not a prediction interval for Jill's actual future wage path. The ratio and its interval rely on the delta-method approximation; using a $t$ critical value does not make nonlinear-function inference exact in finite samples.